# THE TRANSMISSION OF SON PREFERENCE*

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**Abstract**

This paper examines how culturally rooted son preference contributes to missing women. I exploit the quasi-random settlement of mainland Chinese migrants in Taiwan and measure their cultural backgrounds using ancestor worship in their places of origin. Local families exposed to migrants from origins with stronger ancestor worship are more likely to continue childbearing without a son and, after the Legalization of Abortion in 1985, to select sons at higher birth orders. Among second-generation migrants, sex selection is associated not only with the ancestor worship of their fathers' places of origin but also with those of other migrant groups in their communities. The evidence shows how a preference for sons is sustained across generations and spreads beyond the families that pass it down, contributing to missing women.

**Keywords:** Cultural Transmission, Migration, Missing Women, Son Preference.
**JEL codes:** J13, J16, N35, Z10.

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# 1 Introduction

Economic incentives and culture offer complementary explanations for missing women. Increases in women's earnings relative to men's improve girls' survival (Qian, 2008). Cultural beliefs can also encourage son preference by assigning sons special roles in continuing family lines and performing religious rituals (Jayachandran, 2015). Identifying the cultural mechanisms contributing to missing women is challenging because culture is often intertwined with the economic incentives parents face.

This paper provides evidence on a cultural mechanism contributing to missing women: the transmission of son preference within families and across communities. Cultural transmission models describe how preferences are acquired through both parental socialization and interactions outside the family (Bisin and Verdier, 2023). Distinguishing family and social influences is difficult because family background and social environment are closely related. Migration brings people with different cultural backgrounds into a shared setting, but migrants' choices of where to settle can correlate cultural exposure with other local characteristics. I exploit the state-directed settlement of mainland Chinese migrants in Taiwan following the Kuomintang (KMT) Retreat to address this challenge. Using regional variation in ancestor worship as a proxy for son preference, I examine how migrants' ancestor worship reshapes local families' fertility and sex selection. I also examine the transmission of son preference through paternal lineage and between migrant groups within communities by looking at the second-generation migrants.

The KMT Retreat brought about one million mainland Chinese migrants to Taiwan between 1945 and 1954, accounting for roughly 15 percent of the island's population in the early 1950s.[1] The KMT directed the settlement of migrants across Taiwan, limiting their ability to choose destinations. Migrants came from different parts of mainland China and brought distinct cultural traditions to the places where they settled. Their allocation across towns generated variation in the cultural backgrounds of the migrants encountered by local Taiwanese.

I measure migrants' cultural backgrounds using ancestor worship, a tradition in which male descendants continue the family line and perform ancestral rites.[2] I digitize festival records from 2,467 county gazetteers describing people's cultural practices in nineteenth- and early twentieth-century China. For each county, I calculate the share of festival days involving

[1] Previous studies examine how the resulting surplus of men affected women's bargaining power (Francis, 2011), men's entrepreneurship (Chang and Zhang, 2015), and women's late-life mortality (Chang, Kan and Zhang, 2024).

[2] Survey evidence from mainland China links ancestor worship and patrilineal family practice to son preference (Li and Lavely, 2003; Murphy, Tao and Lu, 2011; Hu and Tian, 2018).

ancestor worship. I aggregate these county measures to the 48 mainland origins recorded in the 1956 Taiwan census and weight them by each origin's share of a town's migrant population. The resulting measure, ancestor worship (migrants), captures the prevalence of ancestor worship in the places from which a town's migrants originated.

Identification requires migrants' ancestor worship is unrelated to pre-existing local characteristics that could affect subsequent changes in sex selection. Historical settlement accounts and balance tests provide evidence relevant to this assumption. Neither the male share among children aged 0–10 in 1920 nor local ancestor worship measured using ancestry composition in 1926 predicts subsequent exposure to migrants' ancestor worship. I also examine a broader set of historical characteristics. These results provide little evidence that the allocation of migrant origins across towns was systematically related to pre-existing determinants of local son preference.

The expression of son preference depends on parents' ability to act on it. During the mid-1980s, increasing demand for safe abortions led Taiwanese lawmakers to legalize abortion for women to induce an abortion up to the $24^{th}$ week of pregnancy. The 1985 Legalization of Abortion and ultrasound technology combined to create an efficient and legal method for parents to choose the sex of their children at birth. This change provides an opportunity to examine whether local families exposed to migrants with stronger son preference responded more strongly when prenatal sex selection became more accessible.

I first examine the transmission of son preference from migrants to locals. The empirical strategy is based on a demographic regularity: the sex ratio of the first two births is biologically normal but becomes abnormally male-biased at the third and higher birth orders (Lin and Luoh, 2008). I find that, after 1985, one residual standard deviation (s.d.) increase in ancestor worship (migrants) raises the probability that local Taiwanese parents have a male child at the third and higher birth orders by 0.43 percentage points (p.p.). The event study shows no clear differential trend before the Legalization of Abortion, followed by positive estimates afterward. I find no corresponding differential response at the first two birth orders. Among higher-order births, the point estimate is larger for parents whose first two children were daughters, consistent with a stronger incentive to select for a son when they have not yet had one. The result is robust to alternative specifications and measures of ancestor worship, as well as controls for other festival practices and cultural traits, and sensitivity checks using towns' socioeconomic conditions in 1980. Because the census observes surviving children, I also examine children of comparable ages in the 1980 census, before the Legalization of Abortion. I find no similar relationship between migrants' ancestor worship and the age pattern of children's male shares. This check provides evidence against a pre-existing survival pattern as an explanation for the baseline result.

Son preference can also lead parents to continue childbearing when they have not yet had a son. I find that exposure to migrants' ancestor worship widens the gap in third-birth probabilities between parents with two daughters and those with two sons. This finding complements the sex selection results by linking the same cultural exposure to fertility decisions through which son preference can be expressed. Survey evidence further links migrants' ancestor worship to stronger stated son preference among cohorts after KMT Retreat relative to earlier cohorts within local Taiwanese. These findings connect cultural exposure to both the importance attached to having son(s) and the reproductive decisions through which parents act on that preference.

I next examine how migrants may have influenced local son preference by studying migrants' presence among teachers and peers and the cultural distance between migrants and locals. The effect is stronger in towns with a higher migrant teacher share, consistent with schooling providing opportunities for cultural transmission. Point estimates are also larger in towns with a higher migrant peer share, shorter distances to historical Confucian schools, and higher foot-binding rates in 1920. However, these differences are imprecisely estimated. The teacher result provides the clearest evidence of heterogeneity, while the other patterns remain suggestive.

I next examine how son preference is transmitted among migrants' descendants using an epidemiological approach that compares individuals with different origins living in a shared destination (Fernández and Fogli, 2009; Fernández, 2011). The epidemiological approach compares second-generation migrants whose parents were born in different places in mainland China and thus experienced different levels of ancestor worship, but who now live in the same places in Taiwan and therefore face similar social and local institutional constraints when making decisions. I find that among second-generation migrant fathers, one residual s.d. increase in ancestor worship is associated with a 3.21 p.p. larger post-1985 increase in the probability of a male child at third and higher birth orders. The specification absorbs town-by-birthyear effects, allowing comparisons within a common local environment. The corresponding estimate based on migrant mothers' origins is smaller and imprecise. I then examine whether son preference is also transmitted through migrant communities beyond the family. For second-generation migrants, I measure exposure to other migrant groups in the same town using the average ancestor worship of migrants from origins other than their own. One residual s.d. increase in this measure is associated with a 0.94 p.p. larger post-1985 increase in the probability of a male child at third and higher birth orders.

This paper contributes to three main bodies of literature. First, it provides evidence on a cultural mechanism contributing to missing women. Existing research mainly examines how economic incentives and opportunities for sex selection affect outcomes for girls. Qian

(2008) shows that increases in women's earnings relative to men's improve girls' survival and educational attainment. Almond, Li and Zhang (2019) show that land reform increased sex selection among families whose first child was a daughter, while Guo, Huang and Wang (2025) find that public pensions make the sex ratio at birth more balanced. Other studies examine how the costs of additional births and access to reproductive technologies shape sex selection. Ebenstein (2010) finds that higher fines for unauthorized births discourage fertility and raise more male-biased sex ratios. Ultrasound technology and access to abortion facilitate prenatal sex selection, particularly at higher birth orders (Arnold, Kishor and Roy, 2002; Chen, Li and Meng, 2013; Lin, Liu and Qian, 2014). Evidence from Asian immigrants in the United States and Canada further shows that son preference can persist after migration (Abrevaya, 2009; Almond, Edlund and Milligan, 2013). I extend this literature by exploiting quasi-random migrant settlement to identify how cultural exposure affects reproductive behavior among the receiving population. Exposure to migrants from origins with stronger ancestor worship strengthens local parents' tendency to continue childbearing without a son and increases sex selection at higher birth orders after the Legalization of Abortion. These findings provide evidence on the cultural influences underlying the demand for sons.

Second, the paper contributes to research on cultural persistence and transmission (Giuliano and Nunn, 2021). Studies of migrants and their descendants document the persistence of origin-country differences in women's work and fertility (Blau, Kahn and Papps, 2011; Fernández and Fogli, 2009), often using the epidemiological approach concluded by Fernández (2011). A related literature examines how inter-group contact shapes social attitudes and political preferences, using evidence from pilgrimage and historical immigration (Clingingsmith, Khwaja and Kremer, 2009; Giuliano and Tabellini, 2020). Studies of forced displacement document the diffusion of gender norms to receiving populations (Miho, Jarotschkin and Zhuravskaya, 2024; Jessen, Schmitz and Weinhardt, 2024). Evidence from World War I further shows how changes in women's work can persist across generations and diffuse through social interactions (Gay, 2023; Aneja, Farina and Xu, 2026). I bring these strands together by examining the persistence and diffusion of son preference within the same historical setting. The evidence traces its transmission through paternal lineage and between migrant groups within communities, as well as its spread to local families. These findings show how the cultural mechanisms contributing to missing women extend beyond transmission within families to social interactions across groups.

Third, the paper relates to research on how historical legacies shape contemporary economic outcomes (Acemoglu, Johnson and Robinson, 2005), with a focus on gender norms and reproductive behavior. Historical production systems can leave lasting differences in economic opportunities and gender roles. Michalopoulos, Putterman and Weil (2019) link an-

cestral reliance on agriculture to contemporary education and wealth. Alesina, Giuliano and Nunn (2013) connect historical plough use to less egalitarian gender roles, including among migrants' descendants, while Xue (2016) links women's historical work in cotton weaving to more egalitarian beliefs about women's capabilities. Historical demographic shocks also have lasting consequences for women's work and gender norms, as shown by evidence from colonial Australia (Grosjean and Khattar, 2019) and the transatlantic slave trade (Teso, 2019). Michalopoulos and Xue (2021) measure gender norms through portrayals of men and women in folk tales. I use festival records in gazetteers to measure ancestor worship in migrants' places of origin and link this measure to fertility and sex selection in Taiwan. This connects historical cultural practices centered on paternal lineage continuity to reproductive decisions in a different social and institutional setting.

The remainder of this paper is organized as follows. Section 2 briefly introduces the historical background of ancestor worship and migration in Taiwan. Section 3 describes the data used in the empirical analysis. Section 4 talks about the variation in ancestor worship (migrants) and transmission from migrants to locals. Section 5 discusses the vertical transmission and transmission within migrant communities. Section 6 presents evidence that ancestor worship (migrants) reshapes individuals' beliefs in Taiwan. Finally, Section 7 concludes.

## 2 Historical Background

Mainland China and Taiwan have long been linked by migration and shared cultural practices, including ancestor worship. After Japanese rule ended in 1945, the Chinese Civil War and the KMT Retreat brought a new wave of mainland migrants to Taiwan. This section describes the relationship between ancestor worship and son preference and the historical circumstances of migrant settlement.

### 2.1 Ancestor Worship

#### 2.1.1 Ancestor Worship and Son Preference

China has a long history of son preference. One cultural basis for this preference is ancestor worship, which links the welfare of the living to that of the dead. Ancestors are believed to influence their descendants' fortunes, while depending on offerings from the living for their welfare in the afterlife. Neglecting these obligations is believed to bring misfortune. Because male descendants are expected to worship ancestors, having no sons raises the fear of becoming a "hungry ghost," with no one to provide offerings after death (Baker, 1979; Yang,

1967). Men who share a paternal ancestor and surname gather during festivals and important lineage occasions to "feed" their ancestors. These communal rituals include offering sacrifices, burning joss paper, and tending ancestral graves.

These obligations were closely connected to duties toward living family members (Twitchett, Fairbank and Feuerwerker, 1978). Sons carried the family name and continued the paternal line, while daughters generally married into their husbands' families.[3] Sons and their wives were expected to care for the sons' parents in old age.[4] These arrangements connected the value placed on sons to both support during life and ancestral obligations after death.

Confucian ethics reinforced these obligations through its emphasis on filial piety and family continuity (Zhuo, 2012). Confucius urged followers to sacrifice to their ancestors "as if present." Mencius, the second sage relative to Confucius, described having no posterity as the gravest failure of filial duty. Within a patrilineal family system, this injunction placed particular importance on having son(s).[5] These expectations formed part of a broader family hierarchy that assigned authority to fathers, husbands, and sons, a relationship between Confucianism and gender inequality also discussed by Jia and Kung (2025).[6]

Ancestor worship faced state suppression in mainland China after 1949, particularly during the Cultural Revolution, when ancestral halls were destroyed and rituals were prohibited. The practice gradually revived after the Cultural Revolution. Drawing on survey evidence, Yang and Hu (2012) estimate that 182 million Chinese adults held ancestor worship beliefs or engaged in related practices.

### 2.1.2 Ancestor Worship in Taiwan

Ancestor worship was already established in Taiwan before KMT Retreat. Most local Taiwanese traced their ancestry to Fujian and Guangdong, bringing these practices into family and community life on the island.[7] During Japanese colonial rule, ancestor worship practices were strictly limited, particularly under the assimilation campaigns of the late colonial period.[8] Colonial authorities promoted Shinto worship and sought to replace ancestral

[3]Patrilineal naming remains common. The *Chinese Names Report: 2020* reports a ratio of 1:12 between newborns receiving their mother's and father's surnames. http://www.gov.cn/fuwu/2021-02/08/content_5585906.htm, accessed September 20, 2021.

[4]The common saying is that "a married daughter is like water spilled on the ground"—one you cannot retrieve.

[5]Discussing ancestor worship, Baker (1979) writes: "The living individual knew that he had a continued existence after death only if he could ensure his own posterity. Hence, the desire for a 'hundred sons and a thousand grandsons' may be seen to be a very understandable one."

[6]The three cardinal guides prescribed the authority of rulers over subjects, fathers over sons, and husbands over wives. The three obediences required women to obey their fathers before marriage, their husbands during marriage, and their sons in widowhood.

[7]Local Taiwanese refers to people who, or whose ancestors, lived in Taiwan before 1945.

[8]According to the *Taiwan Governor-General's Archives*, 71% of Taiwanese could speak Japanese in 1944.

tablets in Taiwanese homes with Shinto talismans, known as *Jingu Taima*. These campaigns included the burning of ancestral tablets (Tsai, 1991).

Japanese assimilation campaigns ended in 1945. The subsequent arrival of mainland migrants brought local families into contact with people from origins extending well beyond Fujian and Guangdong. Interaction between groups exposed local families and migrants to different beliefs and practices surrounding ancestor worship and the continuation of the paternal line. This setting allows me to identify the transmission of son preference from migrants to local families and its contribution to missing women.

## 2.2 Chinese Migrants in Taiwan

### 2.2.1 Three Major Waves of Migrants in Taiwan

There have been three major waves of migration to Taiwan since the 17$^{th}$ century. The first wave occurred under Qing rule (1683-1895). Many peasants from Fujian and Guangdong migrated to Taiwan to work in agriculture. Today, over 95% of local Taiwanese are descendants of first-wave migrants.[9] The second wave occurred under Japanese colonial rule. During this period, migration from mainland China was strictly restricted and controlled. Most migrants were from Japan and Korea, making up about 7% of the total population. After World War II and the surrender of Japan, most Japanese and Koreans left Taiwan. The third wave was the KMT Retreat to Taiwan from 1945 to 1954. The KMT brought roughly one million migrants from mainland China to Taiwan within 10 years. These migrants represented about 15% of Taiwan's total population and are the focus of this research.

### 2.2.2 The KMT Retreat

Immediately after the end of World War II, the KMT and the Chinese Communist Party (CCP) fought a civil war from 1945 to 1949. The former was defeated and retreated to Taiwan. Various studies have suggested that the number of civil war migrants to Taiwan is around one million, while the local population of Taiwan in the 1950s was only 6 million (Barclay, 1954; Jacoby, 1966; Lin, 2002). After 1954, the last retreat of the KMT army in Thailand and Myanmar, there was virtually no immigration to or emigration between the island and mainland China until the late 1980s.

The KMT played a central role in organizing the settlement of mainland migrants across Taiwan. Migrants, including both military and civilian populations, were assigned to destinations through state-organized resettlement, while military dependents' villages and surrounding neighborhoods became important settlement locations. This process substantially

[9]The rest local Taiwanese are indigenous peoples.

limited migrants' ability to choose where to live. Moreover, many migrants initially expected to return to mainland China and, therefore had little reason to sort into destinations based on their long run preferences for settlement in Taiwan (Lin, 2009).

#### 2.2.3 Cultural Transmission between Locals and Migrants

In the 1950s, interactions between migrants and local Taiwanese were also shaped by language differences. Many locals did not speak Mandarin Chinese, although some spoke dialects also used in Fujian and Guangdong. First-generation migrants therefore tended to interact more with other migrants, particularly when they came from linguistically distant regions. At the same time, migrants were disproportionately represented among teachers, bringing them into regular contact with local children. These barriers weakened further for the next generation, as second-generation migrants and local Taiwanese attended the same schools, entered the same workplaces, and interacted in everyday life.

Figure I provides a descriptive view of how differences in son preference between the two groups evolved over time. Migrants consistently had a higher male share among children aged 0-4 than locals. Following the KMT Retreat, however, the male share among local children gradually increased and moved closer to the migrant level.

## 3 Data

### 3.1 Ancestor Worship in China

I construct a measure of ancestor worship from festival descriptions in 2,467 county gazetteers published in the 1980s, which document how local communities celebrated traditional festivals during the Qing dynasty.[10] I classify festival practices involving ancestors or ancestral graves as ancestor worship and measure their prevalence as the share of recorded festival days involving these practices. The construction of the ancestor worship density at the county level is as follows:

$$AW_Density_c = \frac{\#\ of\ days\ of\ ancestor\ worship\ practices}{total\ days\ of\ local\ festivals}$$

Figure II shows substantial regional variation in ancestor worship, with higher values in southern mainland China. Figures A.1 and III show that the rise in the male share across

[10]Many local festivals have the same name but the practices associated with them vary widely by region. For example, within Fujian Province, people in Fuqing County sacrifice deceased ancestors on the winter solstice, but people in Wuping County only make rice wine on the same festival.

birth cohorts was more pronounced in counties with higher ancestor worship density, particularly among cohorts born from the late 1980s onward. This pattern is consistent with ancestor worship sustaining a cultural demand for sons that contributes to missing women. Figure A.2 provides complementary evidence: ancestor worship density is positively associated with the male share among children aged 0–1 in 2000, the number of genealogies, and survey measures of patrilineal and son-preferring norms. These associations further support its use as a proxy for son preference.

## 3.2 Ancestor Worship in Taiwan

I use the 1956 Taiwan census to measure the distribution of migrants from 48 mainland origins across 354 towns. Accordingly, I construct ancestor worship (migrants) as follows:

$$AW_Migrants_a = \frac{\sum_p Migrant_{a,p} * AW_Density_p}{\sum_p Migrant_{a,p}}.$$

where $Migrant_{a,p}$ is the number of migrants from origin $p$ in mainland China living in town $a$ in Taiwan, and $AW_Density_p$ denotes ancestor worship in that origin.[11] Figure IV shows the resulting distribution of ancestor worship (migrants) across Taiwan. The measure captures the ancestor worship represented by migrants from different places of origin in each town.

Most local Taiwanese descended from earlier migrants from Fujian and Guangdong, who brought their ancestor worship traditions to Taiwan before the KMT Retreat. I construct a corresponding measure, ancestor worship (locals), using the *Investigation of Han Ancestral Home in Taiwan*, conducted by Ogawa Naoyoshi in 1926. The survey records, for each town, the number of Han residents tracing their origins to ten prefectures in Fujian and Guangdong.[12] I combine this pre-KMT Retreat distribution of local residents with the same gazetteer-based ancestor worship density used for migrant origins. The resulting measure captures pre-KMT Retreat variation in local ancestor worship traditions. Figure V shows the geographic distribution of ancestor worship (locals), which is higher in western Taiwan, where local Taiwanese families with origins in Fujian and Guangdong were historically more concentrated. By contrast, central and eastern Taiwan are more mountainous and had fewer local Taiwanese communities with such ancestral origins; many residents in these areas were indigenous Taiwanese and therefore less connected to the ancestor worship traditions.

[11] I aggregate the original ancestor worship density from the county level to the historical boundaries corresponding to the census origin categories.

[12] The ten prefectures are Quanzhou, Zhangzhou, Tingzhou, Longyan, Fuzhou, Xinghua, and Yongchun in Fujian Province; and Chaozhou, Jiaying, and Huizhou in Guangdong Province.

I next use ancestor worship (locals) to assess whether ancestor worship captures historical variation in son preference in Taiwan before the KMT Retreat. Figure A.6 shows a strong positive correlation between ancestor worship (locals) and the fraction of males aged 0-10 in 1920. Because both measures predate the post-1945 arrival of mainland Chinese migrants, this relationship provides independent evidence that ancestor worship is associated with historical son preference. This supports the interpretation of ancestor worship (migrants), the main treatment variable, as capturing variation in the son preference brought by mainland Chinese migrants after 1945.

### 3.3 Taiwan Population Census

I use full-count microdata from the 1980 and 1990 Taiwan censuses, covering approximately 18 million and 20 million individuals, respectively. Figure A.3 shows the trend of the fraction of males by birthyear from the 1980 and 1990 censuses. The fraction of males is higher than the biological normal fraction of males at birth (51 percent) in 90% of the cohorts. This indicates a persistent male-biased sex ratio pattern in Taiwan.[13] The figure also reveals two distinct patterns. Before 1985, when the fertility policy environment was relatively stable, the male share increased gradually across birth cohorts. This gradual change motivates the analysis of whether the diffusion of son preference contributed to the evolution of sex ratio. After 1985, the male share rose much more sharply, coinciding with the Legalization of Abortion and the increase in reported abortions shown in Figure VI.[14]

The 1990 census records each individual's relationship to the household head. I use this information to link parents and children in two household arrangements. First, I link children of the household head to the head and the head's spouse. Second, I link grandchildren to their parents only in households with a single co-resident couple among the head's children and their spouses. I exclude observations for which parent-child links cannot be uniquely established. Within each linked family, I reconstruct birth order by ranking co-resident children from oldest to youngest. Figure VII plots the male share by birthyear and reconstructed birth order. The increase after the Legalization of Abortion is more pronounced at third and higher birth orders, consistent with the concentration of prenatal sex selection at these orders documented by Lin, Liu and Qian (2014) using Taiwan's National Birth Registries, reproduced in Figure A.4.

The 1990 census records individuals' registered origin (*jibie*) to distinguish mainland

[13] Infanticide of female children was widespread in ancient China, including Taiwan (Chen, 1989; Lee et al., 1994). Taiwanese people apparently have not practiced infanticide since the twentieth century (Lee, 1981).

[14] The share of abortions is constructed from the 1986 Knowledge, Attitudes, and Practice of Contraception (KAP), the only survey with abortion statistics at the individual-pregnancy level in the 1980s.

migrants and their descendants (*migrants*) from local Taiwanese (*locals*). Registered origin generally follows the paternal line, so this classification includes Taiwan-born descendants of mainland-origin fathers. Information on parents' registered origins allows me to examine cultural transmission associated with parental origins and migrant communities.

**1920 - 1956 Censuses.** To construct explanatory variables and historical controls, I digitize the 1920, 1930, 1940, and 1956 Taiwan censuses. From 1920, 1930, and 1940 censuses, I extract information on ethnic distribution, the fraction of males, and population density. The 1956 census records the residence of migrants at the town level with their origin at the province level in mainland China. It also includes other migrant characteristics such as the migrant rate and the fraction of males among migrants.

## 3.4 Other Controls

In this section, I list the data sources for the set of covariates used in the empirical analysis. I control for factors that may be correlated with son preference.

**Proxies for Confucianism.** I constructed two historical variables, Confucian scholars (*jinshi*) density and Confucian clan density, to capture the influence of Confucianism.[15] Information on Confucian scholars comes from the China Biographical Database Project created by Harvard University.[16] Confucian clan data come from the Comprehensive Catalogue of Chinese Genealogies collected by Yuhua Wang.[17] Historical population data are extracted from Ge (2000). Combining these data sources, I construct Confucian scholars density and Confucian clan density in ancient China.

**Geographic Characteristics.** I collected a broad set of geographic characteristics. I use these variables to test the exogeneity of ancestor worship (migrants) and as controls in regressions. Data on the soil suitability index of wheat, wet rice, maize, white potato, cotton, and tea are from FAO GAEZ (v4.0).[18] I also constructed the logarithmic distance to Taipei (the capital of Taiwan) and the logarithmic distance to the seashore based on the Taiwan administrative map.[19]

**Family Characteristics.** I include a number of family characteristics to control for the family heterogeneity in sex selection. The controls include the years of schooling of the parents, the age of parents at the birth of the child, and the birth order.

[15] *Jinshi* are people who passed national-level bureaucratic examinations. Most political elites in ancient China held the title of *jinshi*.

[16] link:https://projects.iq.harvard.edu/cbdb/home. Last access: 05/08/2022.

[17] link:https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/POOVF6. Last access: 05/08/2022.

[18] link:https://gaez.fao.org/. Last access: 23/01/2022.

[19] link:https://data.gov.tw/dataset/7441. Last access: 23/01/2022.

I report summary statistics of town level characteristics in Table A.1 and individual characteristics from the 1990 census in Table A.2.

# 4 Cultural Transmission from Migrants to Locals

In this section, I first present the main specification of horizontal transmission from migrants to locals, discuss identification assumptions, and present estimation results.

## 4.1 Empirical Strategy

To examine the impact of ancestor worship (migrants) on the sex ratio, I implement a difference-in-differences empirical strategy on the sex of children aged 0-9 in the third and higher birth orders from the 1990 census:

$$\begin{aligned} Male_{ifat} =& \alpha + \beta AW_Migrants_a \times Post_t + \boldsymbol{\omega} \boldsymbol{X_a'} \times Post_t \\ &+ \boldsymbol{\phi} \boldsymbol{X_f'} \times Post_t + \lambda_a + \theta_t + \epsilon_{ifat} \end{aligned} \tag{1}$$

where $Male_{ifat}$ equals one if child $i$, born in year $t$ to family $f$ in town $a$, is male, and zero otherwise. $Post_t$ equals one if the child is born in or after 1985. $\boldsymbol{X_a'}$ is a matrix of town characteristics, including historical controls and geographic characteristics.[20] $\boldsymbol{X_f'}$ is a matrix of family characteristics, including parents' years of schooling, parents' ages at childbirth, and the child's birth order. $\lambda_a$ and $\theta_t$ are town fixed effects and birthyear fixed effects, which capture unobserved town-specific variation and year shocks. Standard errors are clustered at the town level.

## 4.2 Identification Assumptions

The main identification assumption to establish the causal effects of ancestor worship (migrants) is that the destination choices of migrants with different cultures are not affected by local culture and gender norms. In other words, ancestor worship (migrants) should be orthogonal to all observed and unobserved determinants of local son preference. Both the historical narrative and the regression analysis presented below provide strong support for this assumption.

I address the identification challenge in three ways: (1) by presenting the trend of

[20] *Historical Controls:* All the variables listed in Table I, including fraction of males aged 0-10, (ln)distance to Confucian school, foot-binding rate, ancestor worship (locals), fraction of males of locals, population density, fraction of Japanese, fraction of migrants, and fraction of males of migrants. *Geographic characteristics:* the suitability index of cotton, maize, wet rice, white potato, wheat, and tea; (ln)distance to seashore, (ln)distance to Taipei.

the fraction of males at birth before KMT Retreat by the three groups of towns with low, medium, and high levels of ancestor worship (migrants); (2) by regressing ancestor worship (migrants) on a number of cultural and demographic characteristics before KMT Retreat, as well as some migrant characteristics and geographic characteristics; (3) by reporting a balance test of the estimation of historical characteristics with ancestor worship (migrants).

In addition, I show that the number of people who migrated to Taiwan is strongly related to total population and population density of their origins in mainland China. The total population and population density of the origins jointly explain 70.2% of the variance in migrant population in Table A.5. However, it does not exhibit a strong correlation between migrant population and ancestor worship density or any other historical controls in migrants' origins. This evidence supports the argument that migrants were not selected on their origin cultures, as most of them were forced to move to Taiwan under strict military and political orders.

### 4.2.1 Historical Narratives

The assignment of migrants to their destinations was designed by the KMT and guided by the goal of political control over Taiwan. For example, healthy and strong migrant men were assigned to military service or manual labor, and educated individuals were sent to teach Mandarin Chinese. Since Taiwan was under martial law from 1949 to 1987, both the occupations and destinations of migrants were strongly determined by the KMT authority. Historians also argue that, under KMT propaganda and at the beginning of the Korean War, migrants generally believed that KMT will retake the mainland China soon with the help of the U.S. army. As a result, many migrants initially paid little attention to where they settled in Taiwan, only to find themselves unable to return to mainland China for more than four decades.

Before 1945, local Taiwanese settlement patterns were largely shaped by earlier migration waves, Japanese colonial rule, and indigenous settlement. In 1940, about 5% of the population was Japanese, most of whom lived in urban or suburban areas in Taiwan's western plain.[21] About 91% of the population traced their ancestry to Guangdong and Fujian in mainland China, and they were also concentrated in the western plain. The remaining population consisted mainly of indigenous groups living in the eastern mountainous areas. This pre-1945 settlement structure suggests that local demographic and cultural conditions were formed before the arrival of mainland migrants. The key concern is therefore whether migrants from different origins were systematically assigned to towns with different pre-existing

[21] Following Japan's defeat in World War II, most Japanese residents left Taiwan.

demographic, economic, or cultural characteristics. The balance tests below directly examine this possibility.

### 4.2.2 Balance Tests

First, in Figure A.5, I show the relative trends of the fraction of males for local Taiwanese across three groups that experienced high, medium, and low ancestor worship (migrants). Towns with different levels of ancestor worship (migrants) have similar pre-KMT Retreat trends in the fraction of males. This suggests that ancestor worship (migrants) has little correlation with changes in local gender norms prior to the KMT Retreat.

To assess whether the allocation of migrant origins across destinations is correlated with pre-existing local son preference, I regress ancestor worship (migrants) on a set of historical cultural and demographic characteristics, as well as geographic characteristics, and report the estimates in Table I. In the first two columns, the pre-KMT Retreat measure of son preference, the fraction of males aged 0-10 in 1920, has no predictive power. After adding observable historical and geographic characteristics in columns 3 to 6, no observable variable strongly predicts ancestor worship (migrants). Moreover, all controls together explain about 16.3% of the variance in ancestor worship (migrants).[22] Overall, the analysis supports the assumption that variation in ancestor worship (migrants) is largely orthogonal to pre-existing local characteristics, including those related to gender norms and son preference.

I next conduct a balance test to further examine whether ancestor worship (migrants) is systematically correlated with pre-existing local characteristics. Figure VIII shows the estimates for this exercise. The estimates suggest that towns that happen to have higher ancestor worship (migrants) are generally not significantly different from towns that have lower ancestor worship (migrants). These results provide additional support for treating ancestor worship (migrants) as plausibly exogenous to pre-existing local son preference.

## 4.3 Baseline Results

I examine whether exposure to migrants' ancestor worship increased son preference among local Taiwanese parents. The baseline analysis focuses on third and higher order births to local parents who were both born after 1954, the end of the KMT Retreat.[23] If ancestor worship (migrants) strengthened son preference among locals, the Legalization of

[22]Notably, the county fixed effects itself explains about 10% of the variance in the ancestor worship (migrants). All the listed observables together only explain around 6% of the variance.

[23]This restriction ensures that parents were exposed from birth to the post-KMT Retreat migrant environment. It also reduces measurement error arising when older children no longer live with their parents and are therefore missing from the reconstructed birth order and sibling sex composition.

Abortion should lead to a larger increase in male births in towns with greater exposure to this culture.

Table II presents estimates of Specification 1. Column 1 includes town and birthyear fixed effects, which absorb persistent differences across towns and common shocks across birth cohorts. Although migrants had little choice over where they settled, the KMT may have allocated them partly according to local historical and geographic conditions. Columns 2 and 3 successively add historical and geographic characteristics interacted with $Post_t$. These controls allow towns with different historical legacies and geographic conditions to experience different changes in sex selection after Legalization of Abortion. Finally, in column 4, I allow the response to Legalization of Abortion to vary with family characteristics by interacting the child's birth order, parents' ages at childbirth, and parents' years of schooling with $Post_t$. I take this as the baseline specification.

The coefficients on ancestor worship (migrants) interacted with $Post_t$ are stable across columns 1-4, ranging from 0.28 to 0.31, and is statistically significant at the 1% level throughout. The baseline coefficient is 0.29. One residual s.d. increase in ancestor worship (migrants) increases the probability of a male birth by 0.43 p.p. after the Legalization of Abortion. These estimates are consistent with migrants' ancestor worship strengthening son preference among local families, who became better able to act on this preference through prenatal sex selection after abortion was legalized.

A natural implication of son preference is that the effect should depend on the sex composition of previous births. If ancestor worship increases the importance of having at least one son, the effect should be larger among parents who have not yet had one. Columns 5 and 6 of Table II divide the sample according to whether parents had a son in their first two births. The coefficient is 0.40 among parents without a son and 0.24 among those with at least one son. The larger point estimate among parents without a son is consistent with this prediction.

One potential concern is that these results reflect pre-existing trends in sex selection. Earlier family planning policies, for example, could have induced differential changes across towns with different levels of ancestor worship (migrants) before abortion was legalized. To examine this possibility, I re-estimate the baseline specification allowing the coefficient on ancestor worship (migrants) to vary by the child's birthyear, using 1984 as the reference cohort. Figure IX shows no clear differential trend before legalization. The coefficients become positive after 1984 and remain positive across subsequent birth cohorts. The timing of these changes coincides with the Legalization of Abortion.

Although previous studies find that prenatal sex selection is concentrated at third and higher-order births, exposure to migrants' ancestor worship could induce selection at earlier

births as well. This possibility is relevant to the preceding comparison by the sex composition of the first two children: if their sex already reflects selection associated with ancestor worship (migrants), the composition of these groups may itself be affected. I therefore repeat the event study for the first two births. Figure A.7 shows that the coefficients fluctuate around zero and their confidence intervals include zero, providing no evidence of differential sex selection at the first two birth orders.

## 4.4 Alternative Interpretations

The baseline results are consistent with the transmission of son preference from migrants to locals. One concern, however, is that ancestor worship (migrants) captures a broader attachment to traditional culture or other cultural traits that influence son preference. Another is that the observed increase in the male share from census data reflects sex-specific child survival rather than prenatal sex selection. I examine these possibilities below.

### 4.4.1 Ancestor Worship and Traditional Culture

A potential measurement concern is that ancestor worship (migrants) captures general traditional Chinese culture rather than a cultural practice specifically linked to son preference. Places where ancestor worship was more prevalent may also have maintained other traditional practices. If migrants from these places brought a broader set of traditional values that influenced local fertility behavior, the baseline estimate could reflect this broader cultural influence.

To examine this possibility, I look at other types of traditional festivals recorded in the same county gazetteers: festivals with religious practices other than ancestor worship, festivals with marketplace or business activities, and festivals with agricultural activities. These categories capture other aspects of traditional religious, commercial, and agricultural life. I construct these measures using the same gazetteers and migrant-origin weights as for ancestor worship (migrants).

Table A.10 reports the results. Column 1 reproduces the baseline, while columns 2-4 replace ancestor worship (migrants) with each alternative festival measure. I find no evidence that exposure to these other traditional practices increased sex selection among local parents. When all measures enter jointly in column 5, the coefficient on ancestor worship (migrants) remains close to the baseline and statistically significant. This contrast supports the interpretation that ancestor worship captures a dimension of traditional culture specifically related to son preference, rather than a general attachment to other traditional practices.

### 4.4.2 Competing Cultural Explanations

Other cultural traits brought by migrants may influence son preference and be correlated with ancestor worship. I consider clan organization, which emphasizes patrilineal ties and the economic and social roles of male descendants (Kung and Wu, 2025); Confucian scholar density, which captures educational traditions and the influence of male scholarly elites (Chen et al., 2020); tea cultivation, which may shape views about women's economic value (Qian, 2008); and rice and wheat cultivation, which capture differences in collectivism and individualism that may affect adherence to traditional family norms (Talhelm et al., 2014).

Table A.11 reports the results. In column 1, I reproduce the baseline. In columns 2-6, I introduce the alternative measures separately, and in column 7, I include them jointly with ancestor worship (migrants). In the joint specification, I find that exposure to migrants from origins more suitable for tea is associated with less sex selection, while exposure to origins more suitable for rice is associated with more. After controlling for these other cultures, I find an effect of ancestor worship (migrants) consistent with the baseline estimate, suggesting that these competing cultural influences do not explain the increase in sex selection among local parents.

### 4.4.3 Child Mortality

The baseline estimate could also be driven by differences in child mortality. Since the census only observes surviving children, the observed male share depends on both sex selection before birth and survival afterward. Boys and girls may face different mortality risks as they grow older. If these differences also depend on ancestor worship (migrants), the comparison across birth cohorts could partly reflect child survival rather than sex selection.

I examine this concern using the 1980 census. I repeat the event study for children born between 1971 and 1980 and observed in the 1980 census. These children cover the same age range as the 1981-1990 birth cohorts from the 1990 census. Figure A.8 shows that the estimates fluctuate around zero, with no clear differences between younger and older children. This finding suggests that age-related differences in child survival are unlikely to explain the baseline pattern, though the test cannot rule out changes in mortality after legalization.

## 4.5 Robustness

### 4.5.1 Socioeconomic Changes after Migrant Settlement

An important concern is that towns may have developed differently between migrant settlement and Legalization of Abortion. Ancestor worship (migrants) is constructed using

migrant settlement patterns in 1956, while children's sex is observed in the 1990 census. Economic and demographic changes during this interval may have been correlated with ancestor worship (migrants). If these changes also influenced parents' sex selection behaviors, they could partly account for the baseline estimate.

I examine this concern using town level characteristics measured in the 1980 census. In Figure X, I augment the baseline specification with each characteristic separately, interacted with birthyear fixed effects. These interactions allow the relationship between local conditions and sex selection to vary across birth cohorts.[24] First, economic opportunities may affect both the value parents place on sons and their ability to act on these preferences. I therefore consider employment, labor force participation, and average age. I also control for schooling to account for differences in human capital. Marriage markets and family arrangements may also shape fertility and sex selection. I consider marriage timing and marital status, together with the share of women aged 15-35, to capture differences in marriage patterns and the population available for family formation. I also include the share of female as household head as a proxy for women's internal decision making roles. As internal migration may change the composition of local families, I also control for the share of internal migrants. To account for differences in fertility patterns, I control for fertility among women aged 40 and above . Across these specifications, the estimates are similar in both magnitude and statistical significance to the baseline, reported at the top of the figure for comparison.

Second, I consider women's economic and family conditions more directly. Women's employment and education may strengthen their bargaining power within the household and increase the economic value parents place on daughters. Their marriage and family circumstances may also shape fertility decisions. In Figure XI, I therefore control for female-specific economic and demographic characteristics, each introduced separately and interacted with birthyear fixed effects. The estimates are also consistent with the baseline.

Following a similar logic, women's economic opportunities may differ across industries and occupations. I therefore control separately for the female share of employment within each industry and occupation, interacted with birthyear fixed effects. Figures XII and XIII report the respective results. In both figures, the estimates of interest are very similar to the baseline.

[24]These characteristics are measured in 1980, and may themselves have been influenced by cultural exposure. I therefore use them to assess the sensitivity of the estimates rather than include them in the baseline specification.

### 4.5.2 Additional Sensitivity Checks

I conduct several additional checks to examine whether the results are sensitive to the construction of ancestor worship (migrants), the specification, and the choice of standard errors.

First, migrants' mainland origins are observed only at the province level, while ancestor worship is measured at the county level. The baseline measure takes a simple average across counties within each province. This average may imperfectly capture migrants' cultural backgrounds if some counties contributed more migrants than others. I therefore construct alternative provincial averages weighted by county population from the 1953 China Population Census or by county area. I also replace festival shares with counts of ancestor worship festivals, using both unweighted and weighted averages, to examine whether the results depend on the number of other festivals recorded in the gazetteers.

Table A.7 shows that the alternative measures are strongly and positively correlated with the baseline ancestor worship (migrants). In Table A.8, I replace ancestor worship (migrants) with each alternative measure in turn. The results are consistent with the baseline across specifications, suggesting that the baseline finding is robust to the choices of aggregation weights or the ways of construction of ancestor worship measure.

Counties within a province may have differed in their likelihood of contributing migrants following KMT Retreat. I examine this concern by constructing separate measures using counties in KMT- and CCP-controlled areas in June 1946, just before the outbreak of Chinese Civil War. Table A.9 shows that the estimate based on KMT-controlled counties is consistent with the baseline, while the estimate based on CCP-controlled counties is smaller and indistinguishable from zero. I find the same pattern when including both measures jointly. This pattern is consistent with the KMT-led nature of the migration to Taiwan.

Table A.6 examines sensitivity to alternative standard errors and specifications. Column 1 clusters standard errors at the county level, allowing errors to be correlated across towns within each of the 21 counties. Column 2 uses two-way clustering at the town level and birthyear level following Cameron, Gelbach and Miller (2011). The standard errors are similar to the baseline in both cases. Column 3 adds parents' origins-birthyear fixed effects, allowing outcomes across birth cohorts to differ by parental origin. The estimate is again consistent with the baseline. I next consider the timing of Legalization of Abortion and the choice of functional form. Some children born in 1985 were already too far along in gestation when abortion was legalized for their parents to use it for sex selection. Column 4 therefore assigns the 1985 birth cohort to the pre-period. The estimate remains consistent with the baseline. Columns 5 and 6 replace the linear probability model with logit and probit models. The marginal effects, evaluated at the means of the covariates, are close to

the baseline estimate.

## 4.6 Heterogeneous Effects

I next examine whether the effects of migrants' ancestor worship vary with opportunities for contact and cultural similarity between migrants and locals. Cultural values may spread through interactions with peers or through learning from older individuals outside the family, such as teachers (Bisin and Verdier, 2011). The baseline estimates do not distinguish these routes of transmission. I therefore consider the presence of migrant teachers and peers, alongside historical measures of local culture.

After World War II, the departure of Japanese teachers left Taiwan with a severe teacher shortage. The KMT recruited teachers from mainland China and promoted Mandarin as the language of instruction. Migrant teachers consequently occupied an important position in local education, with regular access to students and authority in the classroom. This setting motivates examining cultural transmission through teachers alongside peer interactions. Using the 1980 census, I measure migrant teacher share as the share of migrants among teachers born before 1935, and migrant peer share as the share of migrants among residents born between 1954 and 1974. In the sample, the migrant teacher share averages over 50% and the migrant peer share is around 10%, respectively. Migrants' substantial presence among teachers thus contrasts with their smaller presence among peers. I examine whether the effects of migrants' ancestor worship vary with each share.

Cultural similarity also makes locals more receptive to migrants' values (Spolaore and Wacziarg, 2022; Beach and Hanlon, 2023). I proxy for this similarity using distance to Confucian schools in 1895 and the foot-binding rate in 1920. Proximity to Confucian schools captures familiarity with Confucian traditions, including obligations to ancestors, while foot binding reflects traditional gender norms (Fan and Wu, 2023).

Table A.12 reports estimates of Specification 1 for towns below and above the median of each characteristic in Panels A and B, respectively, with 177 towns in each subsample. Columns 1 and 2 show larger estimates in towns with higher migrant teacher and peer shares. Columns 3 and 4 show larger estimates in towns closer to Confucian schools and with higher historical foot-binding rates. However, the differences between coefficients, reported in Panel C, are statistically significant only for migrant teacher share. This finding is consistent with a role for migrant teachers in cultural transmission, while the patterns for peers and cultural similarity are suggestive.

## 4.7 Parents' Fertility Decisions

Parents who prefer sons may also continue having children until they have a son. This decision matters for interpreting the preceding results, since the sex composition of higher-order births depends on both sex selection and which parents continue childbearing. I therefore examine whether exposure to migrants' ancestor worship strengthens the tendency to have a third child among parents without a son in their first two births.[25] The sample consists of local parents who were both born after 1954 and had at least two children.

Table III reports the results. Column 1 compares parents with no son or one son in their first two births to parents with two sons. Column 2 adds town and parents' birthyear fixed effects, and column 3 further controls for parents' schooling. Across these specifications, parents with two daughters are about 20 p.p. more likely to have a third child than parents with two sons. Parents with one son are about 5 p.p. more likely to do so. Parents with two daughters are more likely to have a third child than parents with two sons. This pattern is consistent with son preference rather than simply a preference for children of both sexes.

In column 4, I interact the indicators for having no son and one son with ancestor worship (migrants). Both interaction coefficients are positive. Greater exposure to migrants' ancestor worship is associated with a larger gap in the third birth probabilities between parents with two daughters and those with two sons. I find the same pattern when comparing parents with one son to those with two sons. These results suggest that migrants' ancestor worship encourages further childbearing to have a son. Columns 5 and 6 divide the sample according to whether the second child was born after 1985. The interaction coefficients are positive in both samples. Greater exposure to migrants' ancestor worship is therefore associated with both more sex selection and a greater tendency to continue childbearing when parents have fewer sons. The fertility results provide additional evidence of son preference beyond the sex composition of higher-order births.

## 4.8 Reverse Transmission: from Locals to Migrants

I also test whether the relationship runs in the opposite direction, from locals to migrants. Table A.13 replaces ancestor worship (migrants) with ancestor worship (locals) and estimates the same baseline specification on the sample of children whose parents are both migrants and born after 1954. Across all four specifications, the coefficients on ancestor worship (locals) interacted with the post-1985 indicator are small and statistically insignificant. The null result suggests that local ancestor worship did not measurably shape migrants' sex selection after the legalization of abortion. This asymmetry supports the interpretation that

[25] Recent work by Chen et al. (2025) shows a similar fertility pattern in Taiwan.

the main results capture cultural transmission from migrants to locals, rather than a general town-level tendency toward son preference.

# 5 Cultural Transmissions within Migrants

In this section, I identify cultural transmission within migrant groups. I focus on second-generation migrants and consider both vertical transmission within migrant families and horizontal transmission within migrant communities.

## 5.1 Epidemiological Approach

I use empirical strategies that mirror the epidemiological approach to cultural transmission proposed by Fernández (2011). Given that the relationship between inherited beliefs and behaviors may be co-determined by institutional factors and the socioeconomic environment, this approach identifies cultural transmission through the behaviors of individuals with different places of origin but who grow up and live in the same environment and thus face similar constraints in their decision-making. Specifically, I exploit variation in sex selection among second-generation migrant parents who make decisions under similar local conditions but whose places of origin differ, and whose migrant neighbors differ in their exposure to ancestor worship.

## 5.2 Transmission through Migrant Fathers

The 1990 census records ancestral origin along the paternal line, allowing me to trace the cultural traditions of second-generation migrant fathers. Table A.3 reports summary statistics for children aged 0-9 at third and higher birth orders whose parents are both second-generation migrants. I examine whether ancestor worship in their ancestral places of origin influences their sex-selection decisions. I estimate the following specification:

$$\begin{aligned} Male_{i,f,j,a,t} =& \beta AW_Density_j \times Post_t + \boldsymbol{\omega_1} \boldsymbol{X_j'} \times Post_t \\ &+ \boldsymbol{\omega_2} \boldsymbol{X_f'} \times Post_t + \sigma_j + \lambda_{k,t} + \delta_{a,t} + \epsilon_{i,f,j,a,t} \end{aligned} \tag{2}$$

where $AW_Density$ is ancestor worship in father $j$'s origin. $\boldsymbol{X_j'}$ contains historical controls from the father's origin: Confucian scholar density, Confucian clan density, and the suitability index of tea, wet rice, and wheat. $\boldsymbol{X_f'}$ is a matrix of family controls containing parents' years of schooling, parents' ages at the birth of the child, birth order, and whether parents are from the same origin. $\sigma_j$ and $\lambda_{k,t}$ are fathers' origin fixed effects and mothers'

origin-birthyear fixed effects, which absorb unobserved effects from the fathers' origin and control for the potential channel from mothers' origins. $\delta_{a,t}$ is town-birthyear fixed effects, which absorb time-varying shocks at the town level, including the impact of horizontal cultural transmission estimated in Section 4. Standard errors are clustered at the fathers' origin level and town level.

Table IV reports the results for cultural transmission through migrant fathers. I restrict the sample to children whose parents are both second-generation migrants, allowing me to distinguish paternal and maternal cultural influences on sex selection. I examine whether migrant fathers from places with stronger ancestor worship became more likely to have sons following the Legalization of Abortion, comparing second-generation migrant families within the same town and birth cohort. The estimates are positive across all three columns. In column 3, the coefficient is 3.28 and statistically significant at the 1 percent level. One residual s.d. increase in ancestor worship in fathers' places of origin is associated with a 3.21 p.p. greater increase in the probability of a male birth following legalization.[26] This finding is consistent with the patrilineal nature of ancestor worship, in which sons carry forward the family lineage and worship traditions. Such traditions may reinforce son preference among migrant fathers and influence their fertility decisions.

I next examine whether a similar pattern exists in migrant mothers. Table A.14 repeats the analysis using ancestor worship in mothers' places of origin. Unlike the paternal estimates, the coefficients are negative and marginally significant in columns 2 and 3. I therefore find no corresponding positive effect through maternal origins. This contrast is consistent with the role of paternal lineage in preserving son preference.

## 5.3 Transmission through Migrant Communities

Under the KMT's settlement arrangements, migrants from different origins often lived in the same communities, bringing diverse cultures into close contact. Daily interactions among these migrants created opportunities for cultural transmission between neighbors. I examine whether the cultures of migrant communities influenced sex selection among second-generation migrants.

I construct a variable indicating that migrants are influenced by their migrant neighbors from other origins within the migrant community. The variable is similar to the previous ancestor worship (migrants) but captures the variation at the individuals' origin-town level.

[26] I use the pre-Legalization of Abortion sample to estimate the residual s.d. of ancestor worship (father).

The variable is constructed:

$$AW_Neighbors_{pi,a} = \frac{\sum_{p \neq pi}(Migrant_{a,p \neq pi} * AW_Density_{p \neq pi})}{\sum_{p \neq pi} Migrant_{a,p \neq pi}}$$

where the migrant from origin $pi$ is influenced by the culture of other migrants from other origins $p \neq pi$ within the migrant community of town $a$ in Taiwan.

To examine community transmission, I compare second-generation migrant fathers from the same origin who live in different towns, where migrant neighbors have different ancestor worship. I estimate a specification similar to Specification 2:

$$Male_{i,f,j,a,t} = \alpha + \beta AW_Neighbors_{j,a} \times Post_t + \boldsymbol{\omega_2 X_f'} \times Post_t + \phi_{f,t} + \delta_{a,t} + \epsilon_{i,f,j,a,t} \quad (3)$$

where $\phi_{f,t}$ absorbs parents' origin-birthyear fixed effects, so the previous channel of transmission within migrant families is absorbed. Standard errors are all clustered at fathers' origin level and town level. $\beta$ is identified from variation in sex selection among migrant fathers from the same origin but surrounded by migrant neighbors from different origins.

Table V reports the results for cultural transmission within migrant communities. The estimates are positive and significant across all three specifications, remaining stable after adding county-father's origin fixed effects and family controls. In column 3, the coefficient is 7.67, implying that one residual s.d. increase in migrant neighbors' ancestor worship increases the probability of a male birth by 0.94 p.p. following the Legalization of Abortion. These findings suggest that the cultural environment in which migrants settled also shaped sex-selection decisions among their descendants, beyond the culture from their own origins.

I repeat the analysis using migrant neighbors from origins different from the mother's origin. Table A.15 reports positive but less precise estimates, with the preferred specification statistically insignificant. Consistent with the previous findings on transmission through migrant fathers, I find no similarly robust positive effect when community exposure is defined relative to mothers' origins.

## 5.4 Comparisons across transmission channels

In order to evaluate the findings of this study on male-biased sex selection and contextualize their magnitudes and impacts, I present the estimates from three distinct cultural transmission channels in Table VI. It is challenging to establish a common framework for comparing the estimated effects across cultural transmission channels, as the samples and sources of variation vary considerably. One potential approach is to compare the estimated effects with a change in one residual standard deviation to compute residual magnitudes.

In other words, one should rely on the effective variation in the treatment variable used for identification once fixed effects are taken into account, and then multiply the raw estimates by these residual standard deviations and relevant sample sizes (Mummolo and Peterson, 2018).[27] To facilitate a more comprehensive comparison of the effects of horizontal and vertical transmission, I compare both estimated coefficients and estimated missing girls, while taking into account the varying group sizes across samples.

The initial focus is on the estimated result, which is based on local parents born after 1954. In comparison to the estimates identified by migrant fathers' origin and migrant fathers' origin-town, the magnitude of the estimate identified on local parents is lower. It appears that the transmission of ancestor worship brought by migrants is more likely to occur horizontally and vertically among migrants than from migrants to locals at the margin. When interpreting estimates with residual standard deviation, the estimate identified by the migrant father's origin has the highest marginal effect, which is consistent with findings in numerous studies indicating that vertical transmission is the most important channel in persisting culture. However, horizontal transmission from migrants to locals has the highest aggregate impact on missing girls because the local population is much larger. One residualized standard deviation increase in ancestor worship (migrants) will lead to 1,122 missing girls, or 3.8 percent of the total missing girls in 1985-90 Taiwan. The comparison between horizontal and vertical transmission indicates that cultural transmission within lineage is the most significant channel in persistence, whereas horizontal transmission is the most significant channel in diffusion. The three channel-specific estimates sum to 1,311 missing girls, equivalent to 4.4 percent of the total estimated missing girls in Taiwan during 1985–1990. s

## 6 Changes in Son Preference

The preceding results show that exposure to migrants' ancestor worship shaped fertility and sex selection among local Taiwanese families. I argue that these responses reflect changes in the importance families attach to having sons. Ancestor worship places particular value on male descendants because they continue the paternal line and fulfill obligations to deceased relatives. Through contact with migrants, local families may place greater weight on these obligations and become more reluctant to remain without a son. Such preferences can influence both whether parents continue childbearing and whether they select the sex of subsequent children. Survey evidence on the importance of having a son provides a way to examine the cultural demand underlying the reproductive decisions documented above.

[27] Gay (2023) applies a similar method to compute standardized estimates and to compare different channels of vertical transmission of female working culture.

This explanation also has implications for how son preference differs across generations. Individuals develop preferences through learning within their families and through interactions with peers and other members of their communities (Bisin and Verdier, 2023). Migrants may consequently influence local families through social contact, while preferences acquired through these interactions may subsequently be passed on to their children. Growing up in communities exposed to stronger ancestor worship could thus reinforce the importance of sons among local Taiwanese without migrant ancestry. I examine this implication by comparing stated son preference among respondents born before 1954 and those born in 1954 or later, relating the difference between these cohorts to ancestor worship (migrants).

## 6.1 Data and Empirical Strategy

I use the 1994, 1999, 2011, and 2016 Taiwan Social Change Surveys. Respondents were asked to evaluate the importance of having at least one son to carry on the ancestral line. The question explicitly connects having a son to patrilineal continuity, making it particularly relevant to the proposed cultural mechanism. I assign 0 to "not important at all" and 1 to "really important," with 0.25-point increments for intermediate responses. The resulting son preference score ranges from 0 to 1, with higher values indicating stronger son preference. I restrict the sample to respondents whose parents are both local Taiwanese, focusing on the diffusion of son preference to the receiving population. The analysis sample contains 6,084 respondents in 128 towns. Summary statistics are reported in Table A.4. The mean of son preference score is 0.52, suggesting that respondents attach moderate importance to having a son.

By 1954, the main wave of migration associated with the KMT Retreat was largely complete. Respondents born in 1954 or later were therefore born into a society where these migrants had already settled. I compare their stated son preference with that of respondents born before 1954 across towns with different levels of ancestor worship (migrants). I estimate the following specification:

$$\begin{aligned} Y_{iast} =& \beta AW_Migrants_a \times After_{it} + \boldsymbol{\omega} \boldsymbol{X_a'} \times After_{it} \\ &+ \boldsymbol{\delta} \boldsymbol{X_i'} + \boldsymbol{\pi} \boldsymbol{X_i'} \times After_{it} + \alpha_a + \sigma_t + \phi_s + \epsilon_{iast}. \end{aligned} \tag{4}$$

$Y_{iast}$ denotes the son preference score of respondent $i$, born in year $t$, living in town $a$, and interviewed in survey wave $s$. $After_{it}$ equals one for respondents born in 1954 or later. $\boldsymbol{X_a'}$ contains historical and geographic town characteristics, and $\boldsymbol{X_i'}$ contains individual controls. Town fixed effects, $\alpha_a$, absorb permanent differences across towns. Birthyear fixed effects, $\sigma_t$, account for differences common to respondents born in the same year. Survey fixed effects,

$\phi_s$, account for common differences across survey waves. Standard errors are all clustered at the town level.

## 6.2 Results

Table VII reports the results. The coefficients on the interaction are all positive and significant at the 1% level across specifications. The results remain consistent across specifications with additional controls and their interactions with $After_i$. In the fully controlled specification, one residual s.d. increase in ancestor worship (migrants) is associated with a 2.24 p.p. larger difference in the son preference score between respondents born after 1954 and those born before.

These results suggest that exposure to migrants' ancestor worship is associated with stronger son preference among the later cohorts relative to the earlier cohorts. The survey evidence links migrants' cultural influence to the importance local Taiwanese attach to having at least a son. Together with the fertility and sex selection results, these findings support a cultural mechanism through which son preference is transmitted and contributes to missing women.

# 7 Conclusion

This paper provides evidence that culture contributes to missing women by shaping the demand for sons. Exploiting the quasi-random settlement of mainland Chinese migrants in Taiwan, I find that exposure to migrants from origins with stronger ancestor worship made local parents more likely to continue childbearing without a son. After the Legalization of Abortion in 1985, greater exposure also increased sex selection at third and higher birth orders. Survey evidence further links this exposure to the importance local Taiwanese attach to having a son to continue the paternal line. Together, these findings connect beliefs about male descendants to the demand for sons and show how this demand shapes different reproductive choices as the options available to parents change.

The evidence on cultural transmission helps explain how this demand for sons persists across generations and spreads beyond the migrant population. Among second-generation migrants, sex selection is associated with ancestor worship in fathers' places of origin, consistent with the importance of paternal lineage in ancestral obligations. Their reproductive decisions are also related to ancestor worship in the origins of other migrant groups within their communities, suggesting that these preferences are sustained through interaction outside the family. The effects on local families show that such influences can cross group boundaries. The stronger effect where migrant teachers are more prevalent is consistent

with schools facilitating the diffusion of son preference. Transmission within families can thus sustain the demand for sons across generations, while social interaction extends its consequences to other families and groups. A migration that many expected to be temporary therefore had demographic consequences that lasted across generations and reached beyond the migrants themselves.

Further research could examine how schools, peers, neighborhoods, and marriages complement or substitute for one another in transmitting son preference across groups. Another direction concerns what happens when contact with the society of origin resumes after a long period of separation. For decades, communication and movement between mainland China and Taiwan were sharply restricted, while political campaigns in mainland China disrupted ancestral practices. A natural next step is to examine whether the beliefs and practices sustained among migrant families and communities in Taiwan later diffused back to mainland China as contact resumed. This would allow me to investigate whether migration can preserve cultural preferences through periods of institutional disruption and whether their subsequent return shapes son preference and reproductive decisions in the society of origin.

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Table I: Ancestor Worship (Migrants) and Historical Characteristics

| Dependent variable | Ancestor worship (migrants) | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Fraction of males aged 0-10 in 1920 | -0.00 | 0.04 | 0.03 | 0.01 | 0.05 | 0.05 |
| | (0.05) | (0.06) | (0.05) | (0.05) | (0.05) | (0.05) |
| (Ln)distance to Confucian school in 1895 | | | -0.10 | -0.07 | -0.04 | -0.06 |
| | | | (0.08) | (0.07) | (0.06) | (0.06) |
| Foot-binding rate in 1920 | | | 0.04 | 0.03 | -0.03 | -0.02 |
| | | | (0.06) | (0.07) | (0.06) | (0.06) |
| Ancestor worship (locals) in 1926 | | | | 0.18 | 0.08 | 0.08 |
| | | | | (0.11) | (0.10) | (0.10) |
| Fraction of males of locals in 1940 | | | | | | -0.08 |
| | | | | | | (0.08) |
| Population density in 1940 | | | | | | 0.03 |
| | | | | | | (0.03) |
| Fraction of Japanese in 1940 | | | | | | -0.05 |
| | | | | | | (0.06) |
| Fraction of migrants in 1956 | | | | | | 0.06 |
| | | | | | | (0.07) |
| Fraction of males of migrants in 1956 | | | | | | 0.06 |
| | | | | | | (0.07) |
| County FEs | No | Yes | Yes | Yes | Yes | Yes |
| Geographic characteristics | No | No | No | No | Yes | Yes |
| # of Towns | 354 | 354 | 354 | 354 | 354 | 354 |
| $R^2$ | 0.000 | 0.100 | 0.105 | 0.126 | 0.149 | 0.163 |

Notes: This table reports standardized coefficients from regressing ancestor worship (migrants) on various demographic characteristics before 1945, migrant characteristics measured in 1956, and geographic characteristics. *Geographic characteristics:* the suitability index of cotton, maize, wet rice, white potato, wheat, and tea; (ln)distance to seashore, (ln)distance to Taipei. Robust standard errors are all clustered at the county level.

Table II: The Effect of Ancestor Worship (Migrants) on 3+ Birth Order Children's Sex

| Dependent Variable | Male = 1 | | | | | |
|---|---|---|---|---|---|---|
| | Full sample | | | | Sons in first two births | |
| | | | | | 0 | $\geq 1$ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Ancestor worship (migrants) X Post | 0.29*** | 0.31*** | 0.28*** | 0.29*** | 0.40*** | 0.24** |
| | (0.10) | (0.08) | (0.08) | (0.08) | (0.12) | (0.11) |
| Birthyear FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| Town FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| Historical controls X Post | No | Yes | Yes | Yes | Yes | Yes |
| Geographic characteristics X Post | No | No | Yes | Yes | Yes | Yes |
| Family controls X Post | No | No | No | Yes | Yes | Yes |
| # of Towns | 354 | 354 | 354 | 354 | 354 | 354 |
| # of Observations | 358,287 | 358,287 | 358,287 | 358,287 | 117,230 | 241,057 |
| Outcome mean | 0.527 | 0.527 | 0.527 | 0.527 | 0.537 | 0.523 |
| $R^2$ | 0.001 | 0.001 | 0.001 | 0.002 | 0.006 | 0.002 |

Notes: This table reports OLS coefficients from estimating Specification (1) on the sample of $3^{rd}$+ birth order aged 0-9 children whose parents are both locals and born after 1954. Columns 5 and 6 restrict the sample to families with no sons and at least one son, respectively, among their first two births. The results from regressing male indicator on birthyear fixed effects, town fixed effects, and a set of historical controls, as well as a series of geographic characteristics and family controls interacting with *Post*. *Geographic characteristics:* the suitability index of cotton, maize, wet rice, white potato, wheat, and tea; (ln)distance to seashore, (ln)distance to Taipei. *Family controls:* birth order of the child, age of parents at child's birth, and years of schooling of parents. Standard errors are all clustered at the town level.

*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table III: Children's Sex Composition and Fertility Decisions for Two-child Parents

| Dependent variable | Third child = 1 | | | | | |
|---|---|---|---|---|---|---|
| $2^{nd}$ birth born after 1985 | All | | | | No | Yes |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| No son in $1^{st}$&$2^{nd}$ birth | 0.21*** | 0.20*** | 0.20*** | 0.07* | 0.13* | 0.04 |
| | (0.00) | (0.00) | (0.00) | (0.04) | (0.07) | (0.03) |
| One son in $1^{st}$&$2^{nd}$ birth | 0.06*** | 0.06*** | 0.05*** | -0.03 | -0.08*** | -0.02 |
| | (0.00) | (0.00) | (0.00) | (0.03) | (0.03) | (0.03) |
| No son in $1^{st}$&$2^{nd}$ birth | | | | 0.43*** | 0.52** | 0.36*** |
| X Ancestor worship (migrants) | | | | (0.14) | (0.24) | (0.11) |
| One son in $1^{st}$&$2^{nd}$ birth | | | | 0.29*** | 0.53*** | 0.19* |
| X Ancestor worship (migrants) | | | | (0.10) | (0.09) | (0.11) |
| Town FEs | No | Yes | Yes | Yes | Yes | Yes |
| Parents' birthyear FEs | No | Yes | Yes | Yes | Yes | Yes |
| Parents' education | No | No | Yes | Yes | Yes | Yes |
| # of Towns | 354 | 354 | 354 | 354 | 354 | 354 |
| # of Observations | 565,324 | 565,324 | 565,324 | 565,324 | 183,463 | 381,861 |
| Outcome mean | 0.383 | 0.383 | 0.383 | 0.383 | 0.638 | 0.260 |
| $R^2$ | 0.025 | 0.091 | 0.133 | 0.133 | 0.118 | 0.075 |

Notes: This table reports OLS coefficients from regressing the indicator of whether parents had a third child on the indicators of having no son and one son in the $1^{st}$&$2^{nd}$ birth, their interactions with ancestor worship (migrants), and years of schooling of parents, with town and parents' birthyear fixed effects. Columns 5 and 6 split the sample by whether the second child was born after 1985. The sample is local parents who had at least two children and were both born after 1954. Standard errors are all clustered at the town level.
*** Significant at the 1 percent level. ** 5 percent level. * 10 percent level.

Table IV: The Effect of Migrant Father's Ancestor Worship on the Sex of 3+ Birth Order Children

| Dependent variable | Male = 1 | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Ancestor worship (father) X Post | 1.88* | 3.12** | 3.28*** |
| | (1.10) | (1.20) | (1.20) |
| Town-birthyear FEs | Yes | Yes | Yes |
| Father's origin FEs | Yes | Yes | Yes |
| Mother's origin-birthyear FEs | Yes | Yes | Yes |
| Origin controls (father) X Post | No | Yes | Yes |
| Family controls X Post | No | No | Yes |
| # of Father's origin | 34 | 34 | 34 |
| # of Towns | 193 | 193 | 193 |
| # of Observations | 5,792 | 5,792 | 5,792 |
| Outcome mean | 0.529 | 0.529 | 0.529 |
| $R^2$ | 0.218 | 0.219 | 0.222 |

Notes: This table reports coefficients from estimating Specification (2) on the sample of $3^{rd}$+ birth order aged 0-9 children whose parents are both second-generation migrants. The results from regressing male indicator on town-birthyear fixed effects, father's origin fixed effects, mother's origin-birthyear fixed effects, and origin controls (father) as well as a series of family controls interacting with *Post*. *Origin controls (father):* Imperial scholars density, Confucian clan density, and suitability index of wheat, wet rice, and tea of father's origin. *Family controls:* birth order of the child, age of parents at child's birth, and years of schooling of parents, as well as whether parents are from the same origin. Standard errors are all clustered at the town level and migrant father's origin level.
*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table V: The Effect of Ancestor Worship of Migrant Neighbors from Origins Other than the Father's Origin on the Sex of 3+ Birth Order Children

| Dependent variable | Male = 1 | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Ancestor worship (neighbors, father) X Post | 7.42** | 7.26** | 7.67** |
| | (3.48) | (3.58) | (3.70) |
| Town-birthyear FEs | Yes | Yes | Yes |
| Parents' origins-birthyear FEs | Yes | Yes | Yes |
| County-Father's origin FEs | No | Yes | Yes |
| Family controls X Post | No | No | Yes |
| # of Father's origin | 26 | 26 | 26 |
| # of Towns | 186 | 186 | 186 |
| # of Observations | 5,601 | 5,601 | 5,601 |
| Outcome mean | 0.527 | 0.527 | 0.527 |
| $R^2$ | 0.239 | 0.287 | 0.291 |

Notes: This table reports coefficients from estimating Specification (3) on the sample of $3^{rd}$+ birth order aged 0-9 children whose parents are both second-generation migrants . The results from regressing male indicator on town-birthyear fixed effects, parents' origins-birthyear fixed effects, and a series of family controls interacting with *Post*. *Family controls:* birth order of the child, age of parents at child's birth, and years of schooling of parents, as well as whether parents are from the same origin. Standard errors are all clustered at the town level and migrant father's origin level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table VI: Estimates' Magnitudes and Estimated Impacts

| | | Ancestor worship ↑ one residual s.d. | | |
|---|---|---|---|---|
| Analysis | Source of variation | Marginal effects | Missing girls | Reference |
| Local parents | town-cohort | 0.43 p.p. | 1122 | Column 4, Table II |
| Second-gen migrant parents | father's origin-cohort | 3.21 p.p. | 147 | Column 3, Table IV |
| Second-gen migrant parents | father's origin-town-cohort | 0.94 p.p. | 42 | Column 3, Table V |

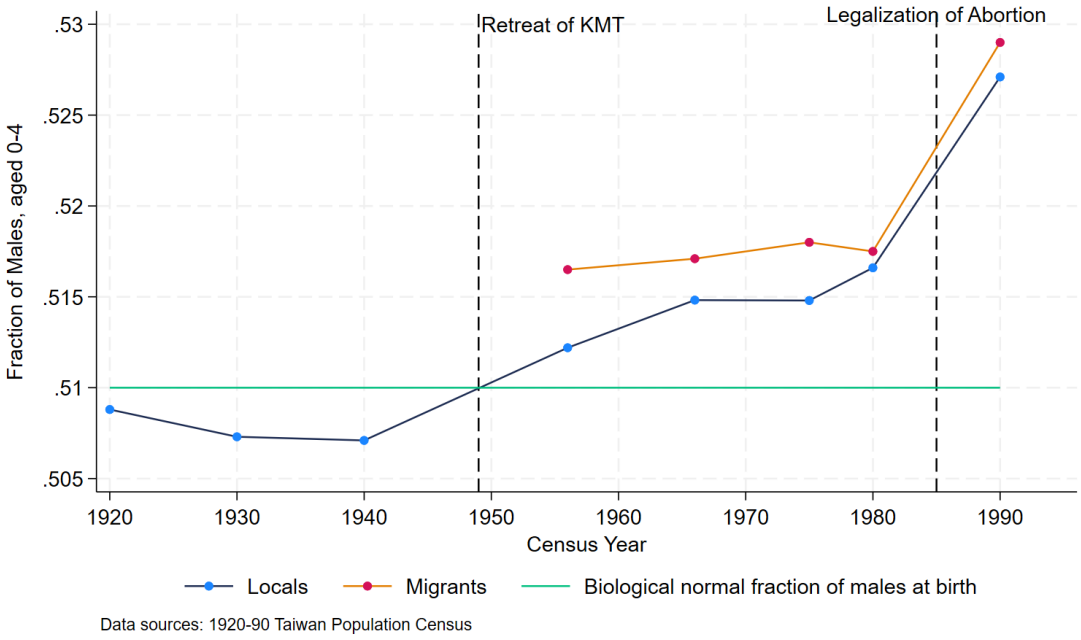
Notes: This table reports the main estimates in this research. *Residual s.d.* is calculated using the pre-1985 sample by residualizing each ancestor worship measure with respect to the relevant controls and fixed effects, excluding those that would fully absorb the measure. *Missing girls* is the estimated number of missing girls by one residual s.d. of the independent variables × the number of children in 1985-1990 of the corresponding samples.

Table VII: Ancestor Worship (Migrants) and Son Preference across Cohorts

| Dependent variable | Son preference score | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Ancestor worship (migrants) × After | 1.05*** | 1.21*** | 1.55*** | 1.82*** |
| | (0.36) | (0.41) | (0.47) | (0.48) |
| Town FEs | Yes | Yes | Yes | Yes |
| Birthyear FEs | Yes | Yes | Yes | Yes |
| Survey FEs | Yes | Yes | Yes | Yes |
| Historical controls × After | No | Yes | Yes | Yes |
| Geographic characteristics × After | No | No | Yes | Yes |
| Individual controls × After | No | No | No | Yes |
| # of Towns | 128 | 128 | 128 | 128 |
| # of Observations | 6,084 | 6,084 | 6,084 | 6,084 |
| $R^2$ | 0.138 | 0.140 | 0.141 | 0.169 |

Notes: This table reports coefficients from estimating Specification (4) using the 1994, 1999, 2011, and 2016 Taiwan Social Change Surveys. The sample consists of respondents whose parents are both local Taiwanese. $After$ equals one for respondents born in 1954 or later and zero for those born before 1954. The outcome ranges from 0 to 1 in 0.25-point increments; higher values indicate greater importance to having at least one son. The results from regressing outcome on birthyear fixed effects, town fixed effects, survey fixed effects, and a set of historical controls, as well as a series of geographic characteristics and individual characteristics interacting with $After$. *Historical controls:* the male share among children aged 0–10 in 1920, log distance to a Confucian school, the foot-binding rate in 1920, ancestor worship (locals), the local male share in 1940, population density in 1940, the Japanese population share in 1940, the male share among migrants, and the migrant population share. *Geographic characteristics:* suitability for cotton, maize, tea, wet rice, white potato, and wheat; log distance to the coast; and log distance to Taipei. *Individual characteristics:* sex, marital status, number of children, and years of schooling. Standard errors are all clustered at the town level.
*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Figure I: Trends of Fraction of Males, aged 0-4




Data sources: 1920-90 Taiwan Population Census

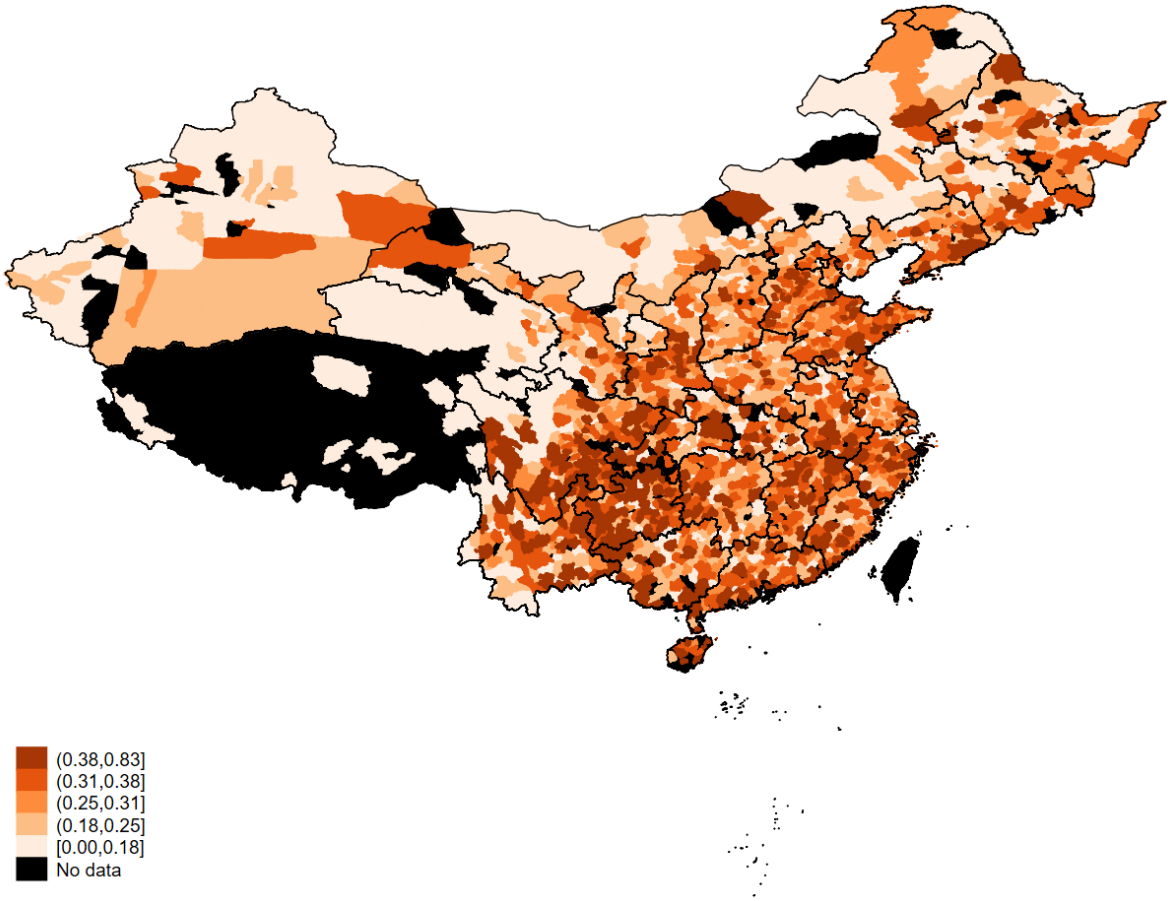
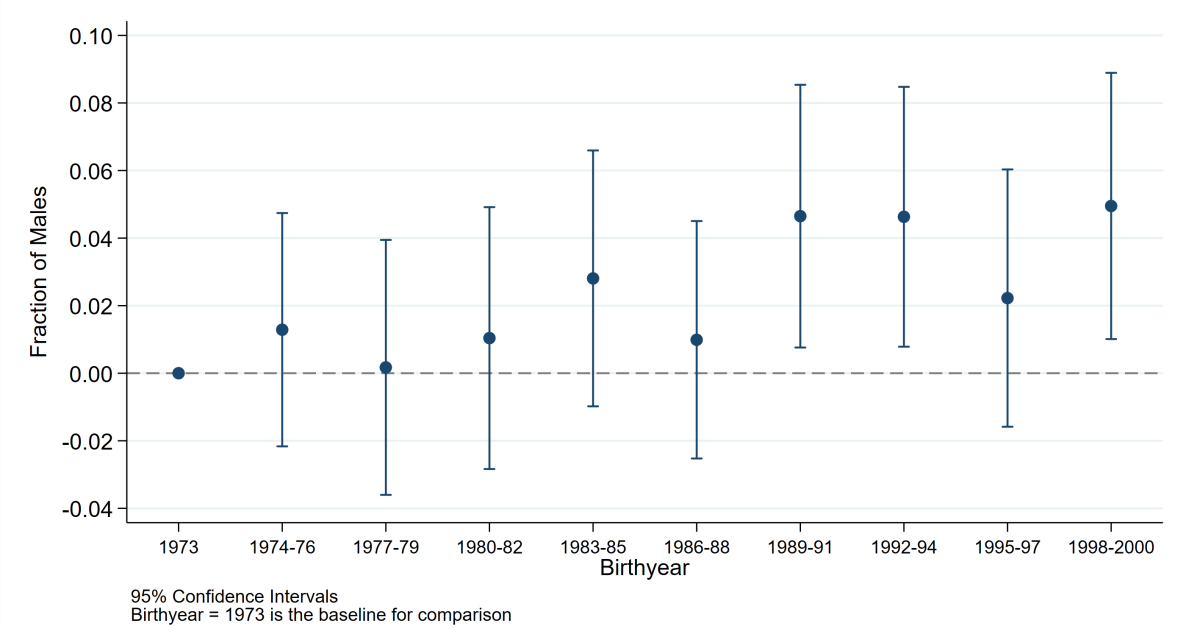



Figure II: The Map of Ancestor Worship Density in China

Figure III: Ancestor Worship Density and Fraction of Males over Time




Note: We regress fraction of males on ancestor worship density interacted with cohort dummies with controlling county fixed effects and birthyear fixed effects. Standard errors are clustered at the county level. We used individuals born in 1973–1982 from the 1% micro sample of 1982 China census, 1983–1990 from the 1% micro sample of 1990 China census, and 1991–2000 from the 1% micro sample of 2000 China census to construct the fraction of males in county-birthyear level.

Figure IV: The Map of Standardized Ancestor Worship (Migrants)



Figure V: The Map of Standardized Ancestor Worship (Locals)

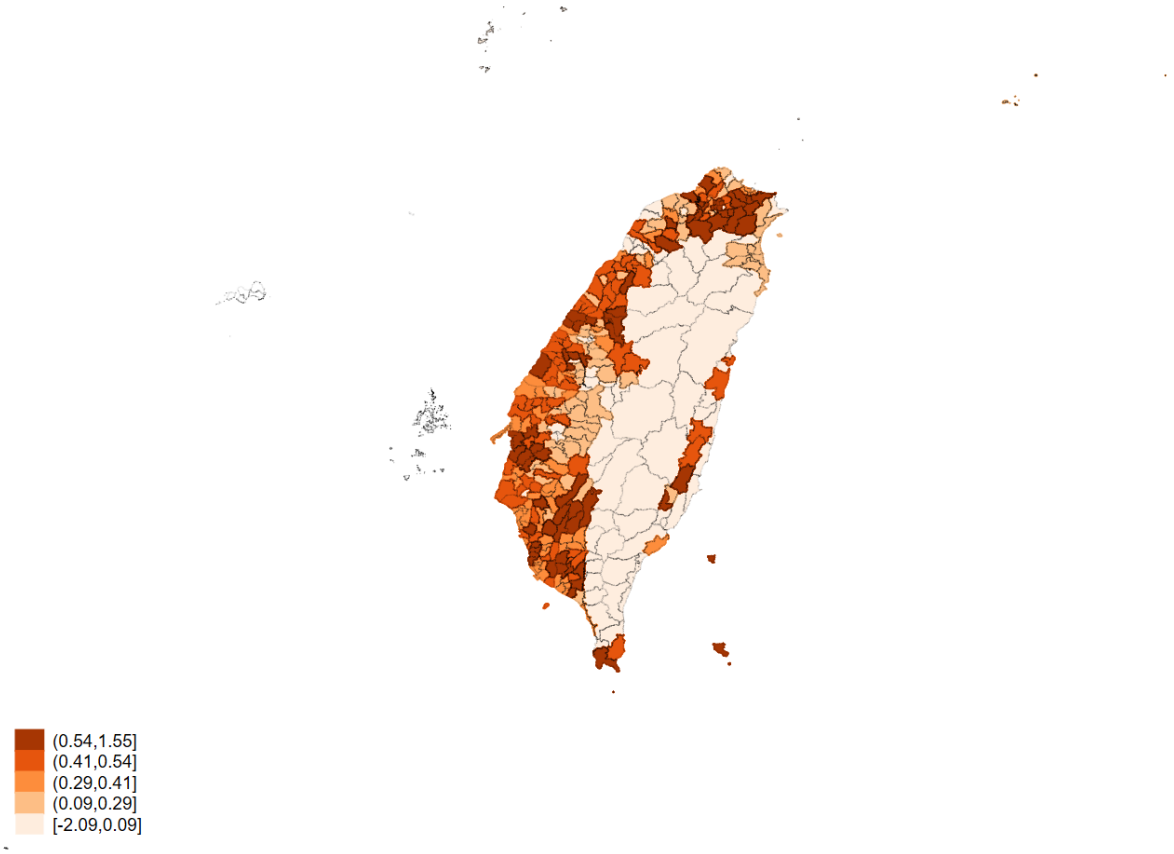
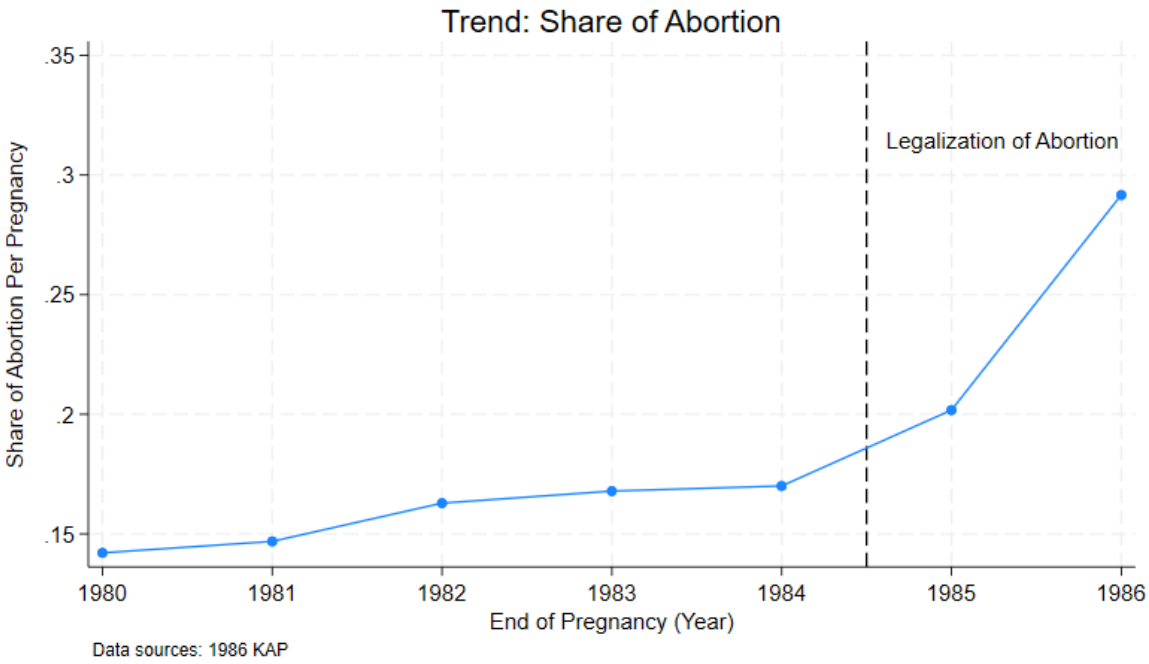
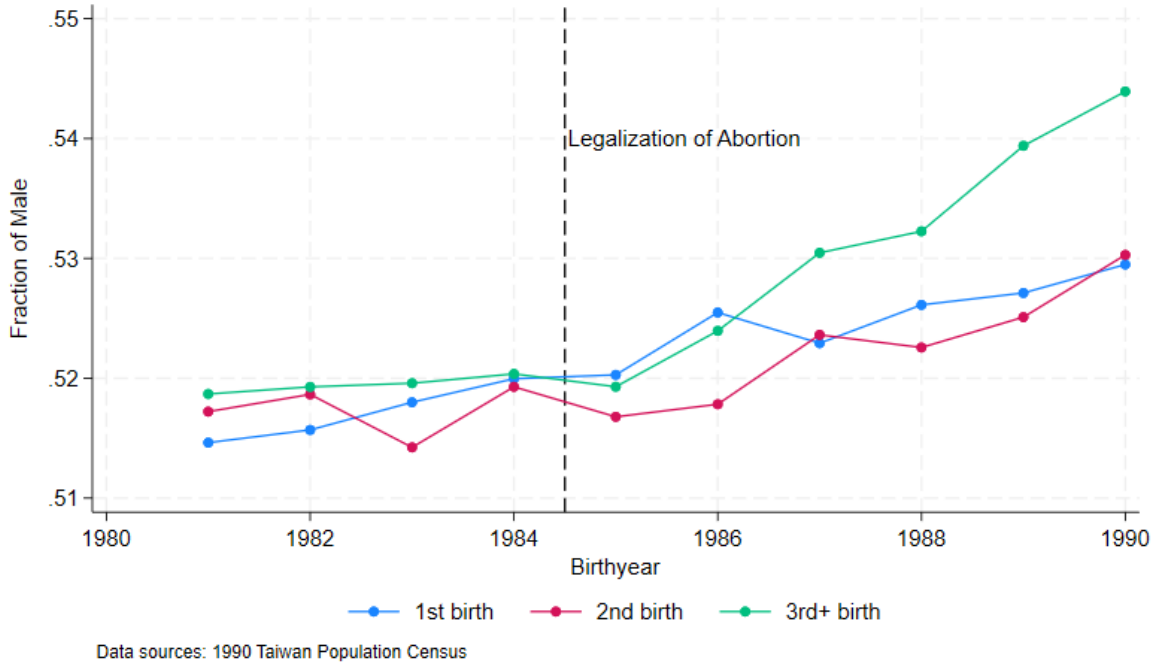


Figure VI: Share of Abortion per Pregnancy (1980-1986)



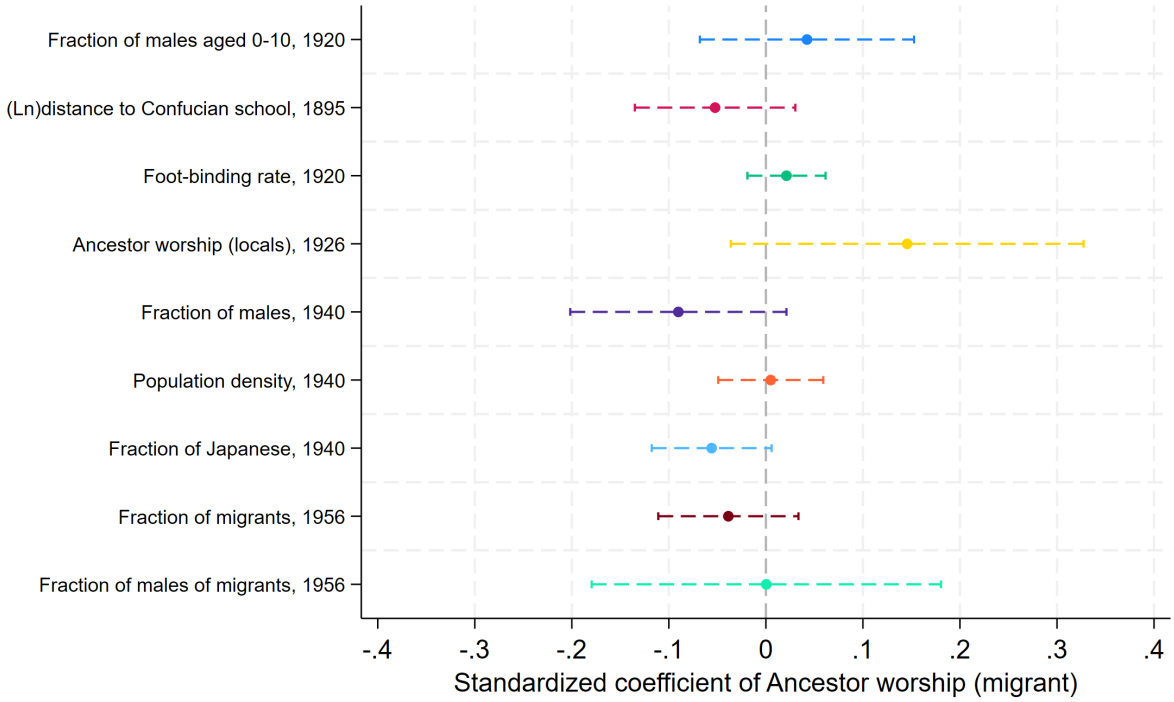

Note: The 1986 KAP is the only survey that contains abortion statistics at the individual-pregnancy level in the 1980s. It surveys married women aged 20-49 and asks the outcomes of every pregnancy.

Figure VII: Fraction of Male by Birth Order (1981-1990)



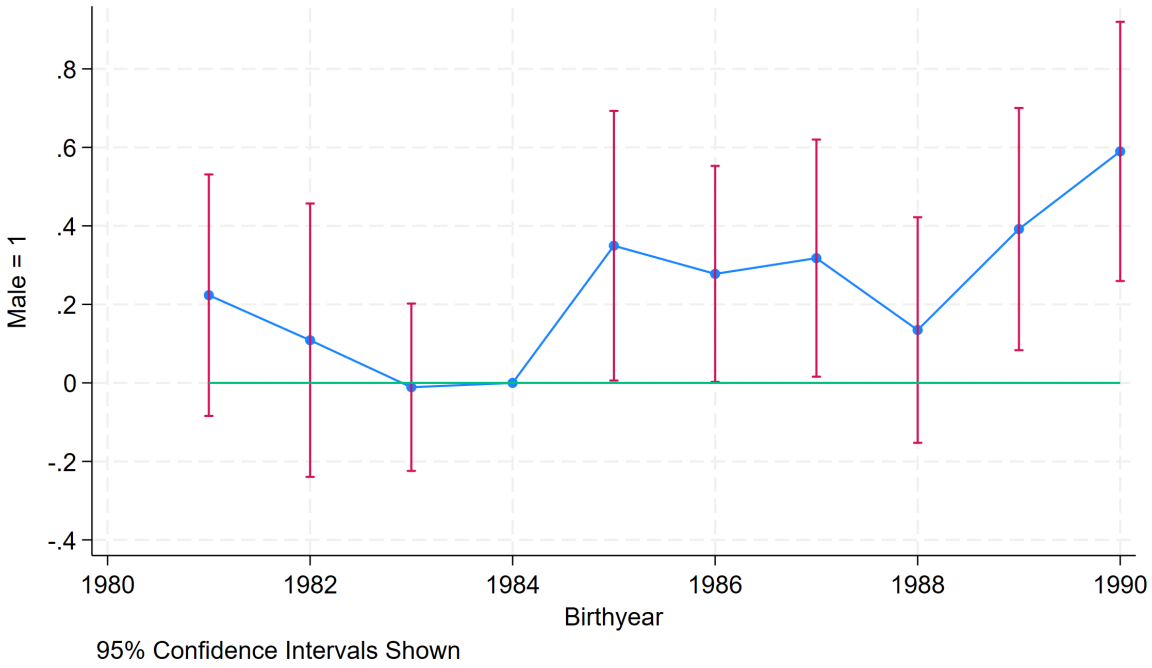

Note: Birth order is constructed based on the family relationship to the household head and the birthyear of each child in 1990.

Figure VIII: Historical Characteristics and Ancestor Worship (Migrant)



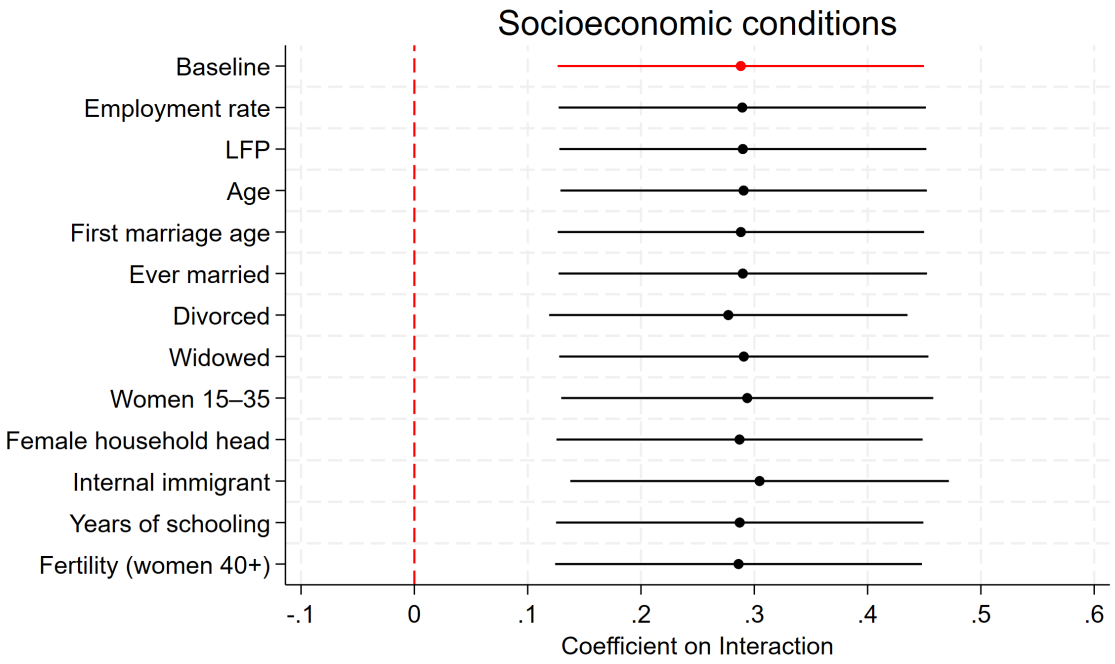

Note: The figure presents the estimated standardized coefficients and respective 95 percent confidence intervals from regressing various historical characteristics (denoted on the y-axis) on ancestor worship (migrants) at the town level, conditional on county fixed effects. All standard errors are clustered at the county level.

Figure IX: Event Study of the Ancestor Worship (Migrants) on Male (1981-1990)



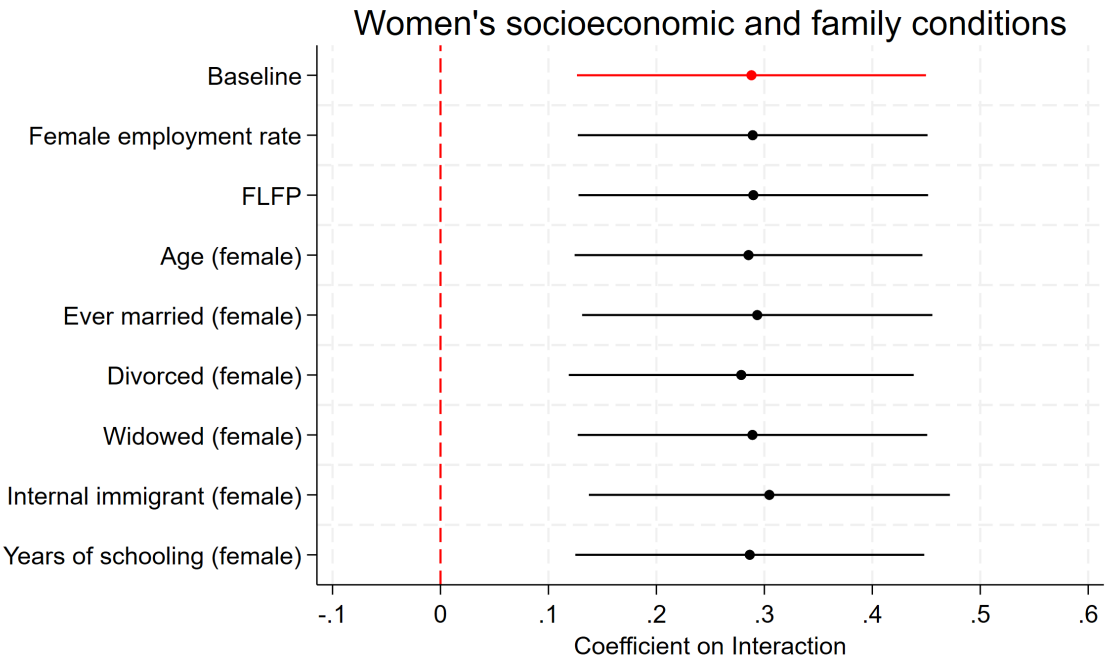

Note: The figure presents the event study by estimating an alternative version of Specification (1), which allows the estimates of ancestor worship (migrants) to vary by birthyear and uses the birthyear=1984 cohort as the comparison group. With this, I can examine pre-trends before the Legalization of Abortion and the dynamic effects of ancestor worship (migrants) after the Legalization of Abortion.

Figure X: The Effect of Ancestor Worship (Migrants) on 3+ Birth Order Children's Sex – Controlling for Socioeconomic Conditions



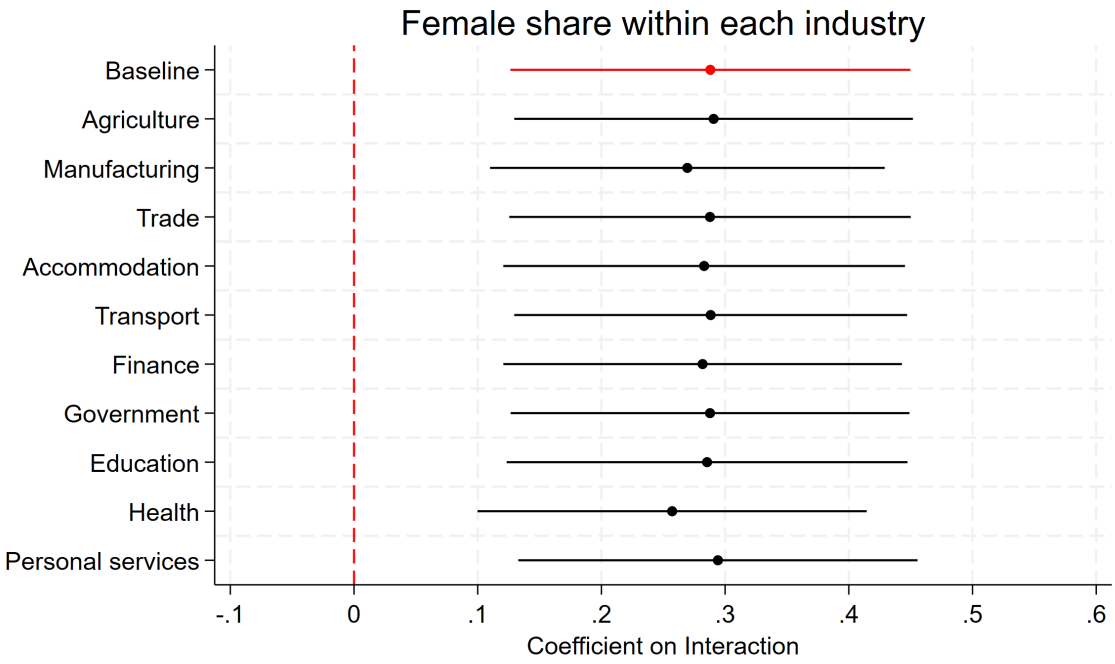

Note: The figure plots coefficients (with corresponding 95% confidence intervals) on the interaction of Ancestor worship (migrants) x Post, augmenting the baseline (in red) with the town controls indicated in each row, interacted with birthyear fixed effects.

Figure XI: The Effect of Ancestor Worship (Migrants) on 3+ Birth Order Children's Sex – Controlling for Women's Socioeconomic and Family Conditions



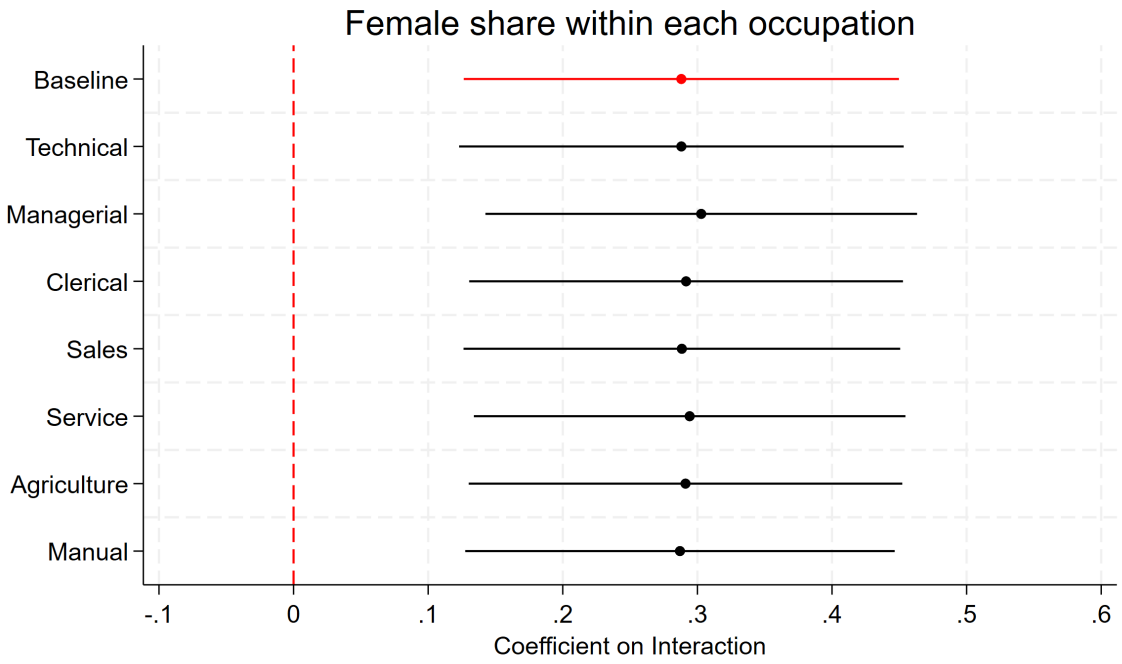

Note: The figure plots coefficients (with corresponding 95% confidence intervals) on the interaction of Ancestor worship (migrants) x Post, augmenting the baseline (in red) with the town controls indicated in each row, interacted with birthyear fixed effects.

Figure XII: The Effect of Ancestor Worship (Migrants) on 3+ Birth Order Children's Sex – Controlling for Female Share within Each Industry




Note: The figure plots coefficients (with corresponding 95% confidence intervals) on the interaction of Ancestor worship (migrants) x Post, augmenting the baseline (in red) with the town controls indicated in each row, interacted with birthyear fixed effects.

Figure XIII: The Effect of Ancestor Worship (Migrants) on 3+ Birth Order Children's Sex – Controlling for Female Share within Each Occupation




Note: The figure plots coefficients (with corresponding 95% confidence intervals) on the interaction of Ancestor worship (migrants) x Post, augmenting the baseline (in red) with the town controls indicated in each row, interacted with birthyear fixed effects.

# Appendix: Figures and Tables

Figure A.1: Fraction of Males in Mainland China

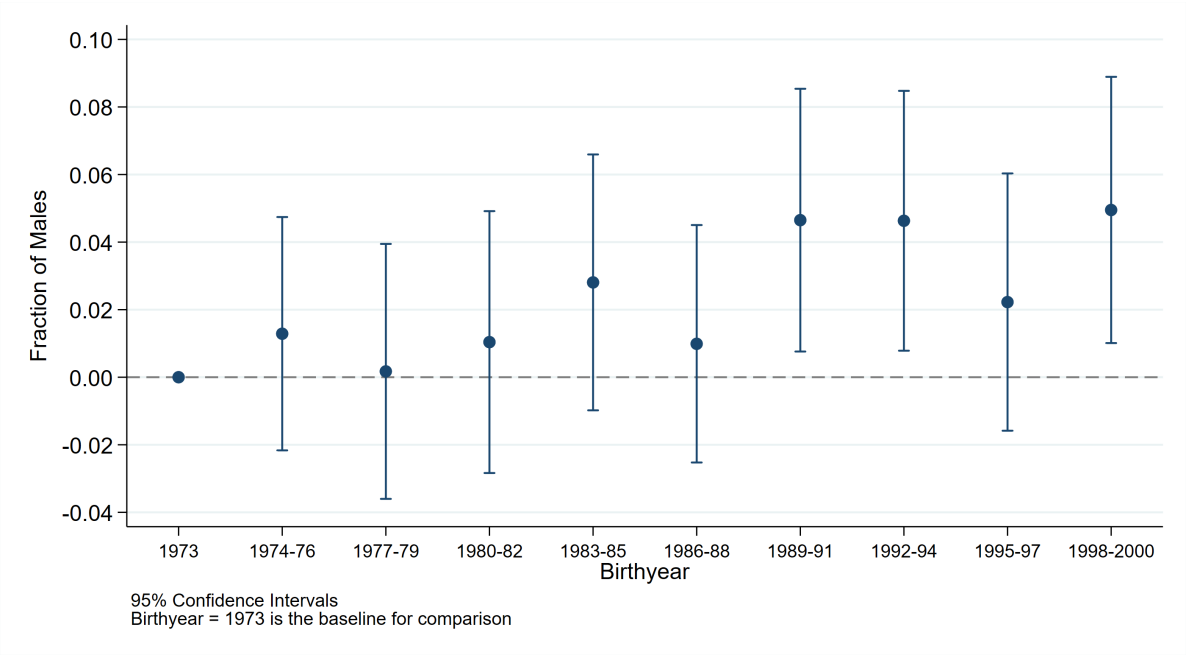


Figure A.2: Correlations between Ancestor Worship and Son Preference

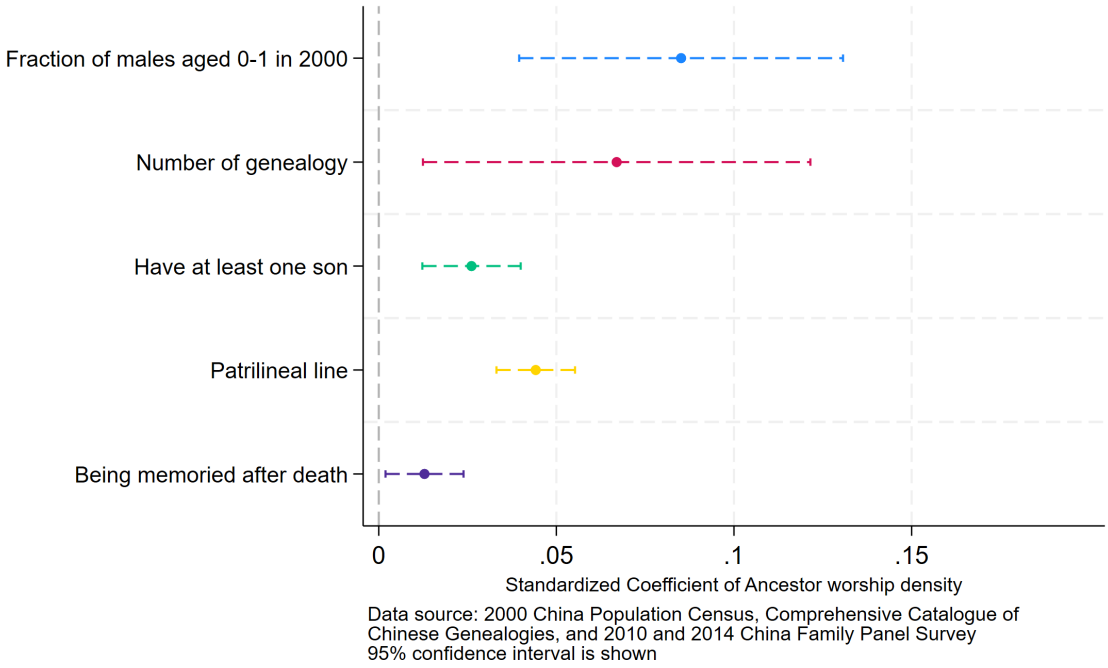
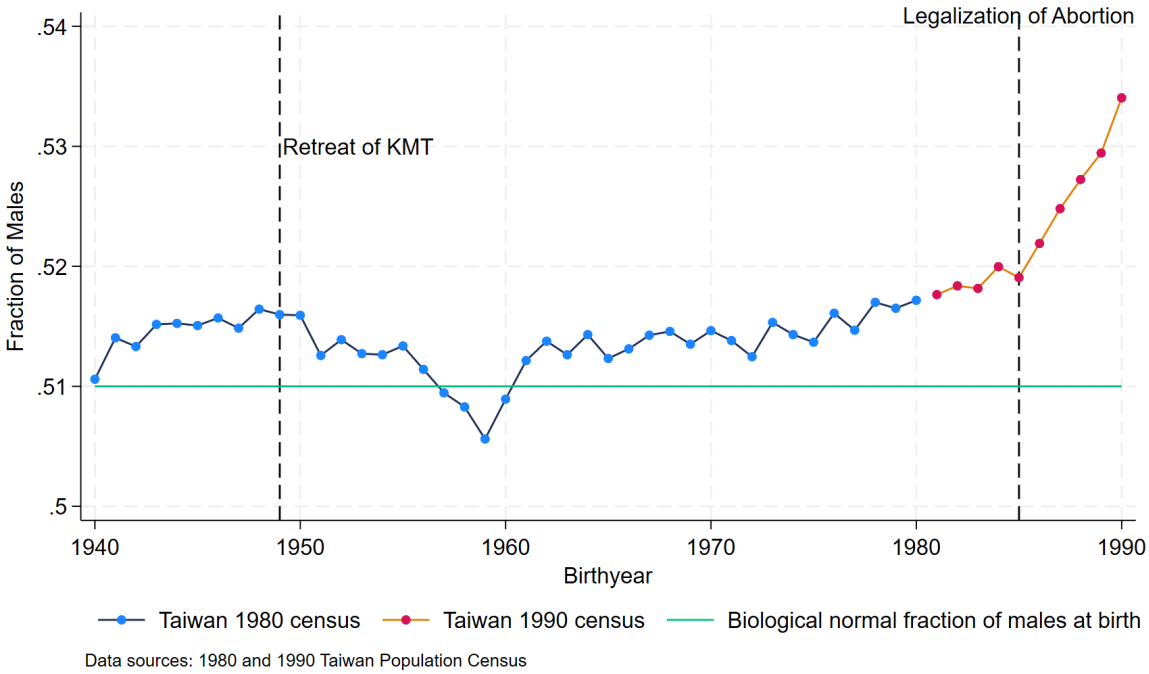


Figure A.3: Fraction of Males in the 1980 and 1990 Censuses




Data sources: 1980 and 1990 Taiwan Population Census

Figure A.4: Fraction of Male of Births by Birth Order (1982-1989) from Lin, Liu and Qian (2014)

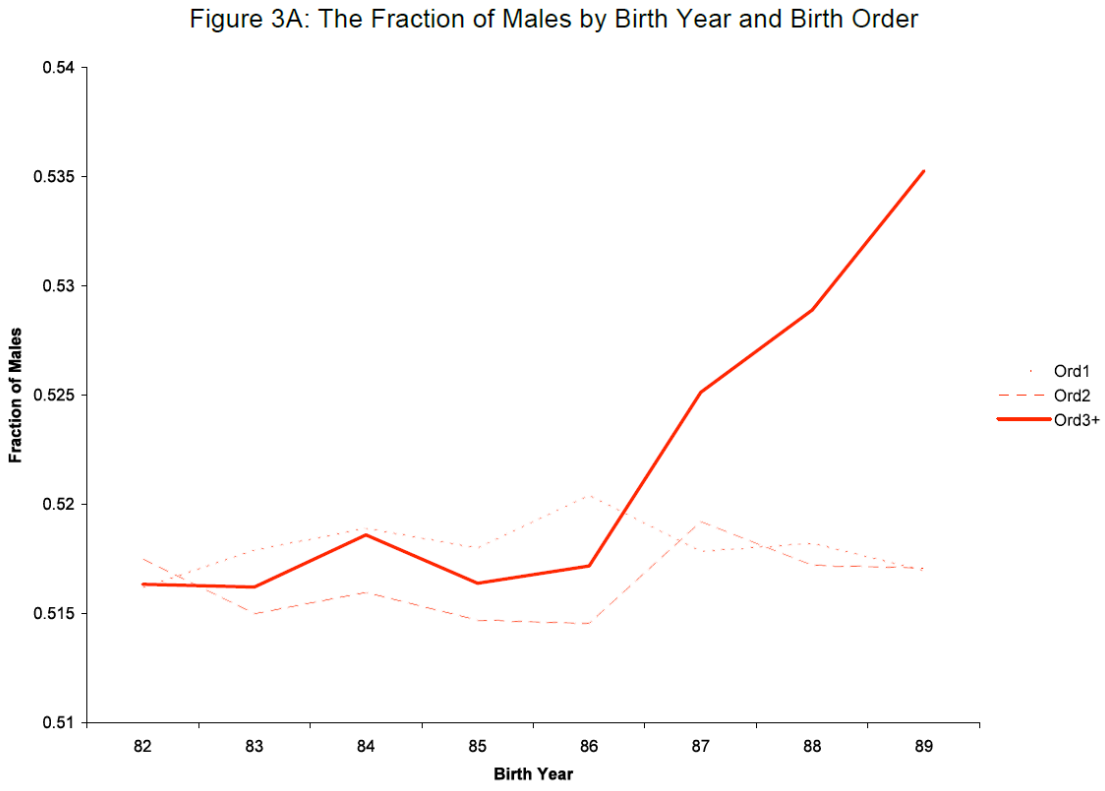
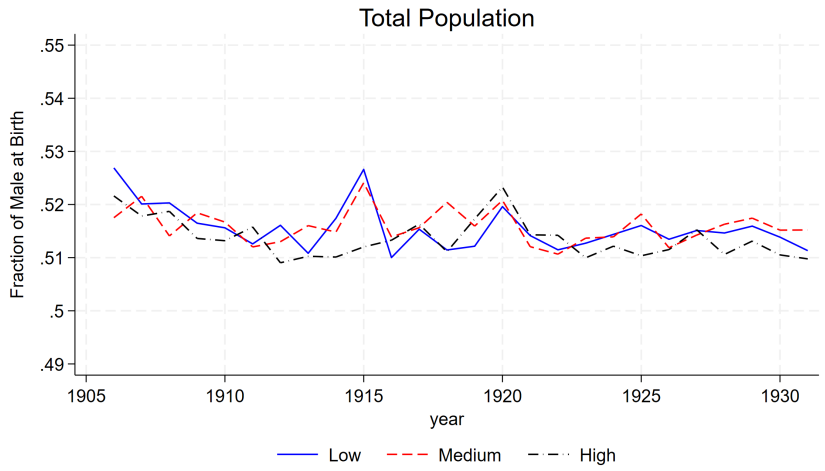
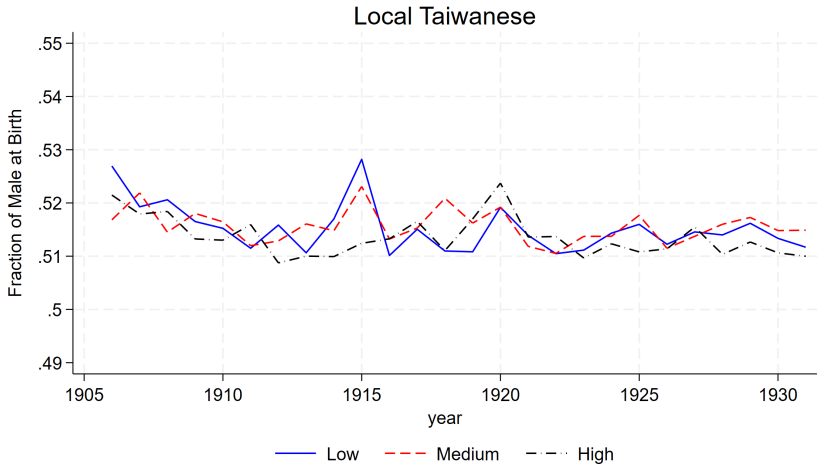


Figure A.5: Trends of Fraction of Males at Birth (1906-1931)



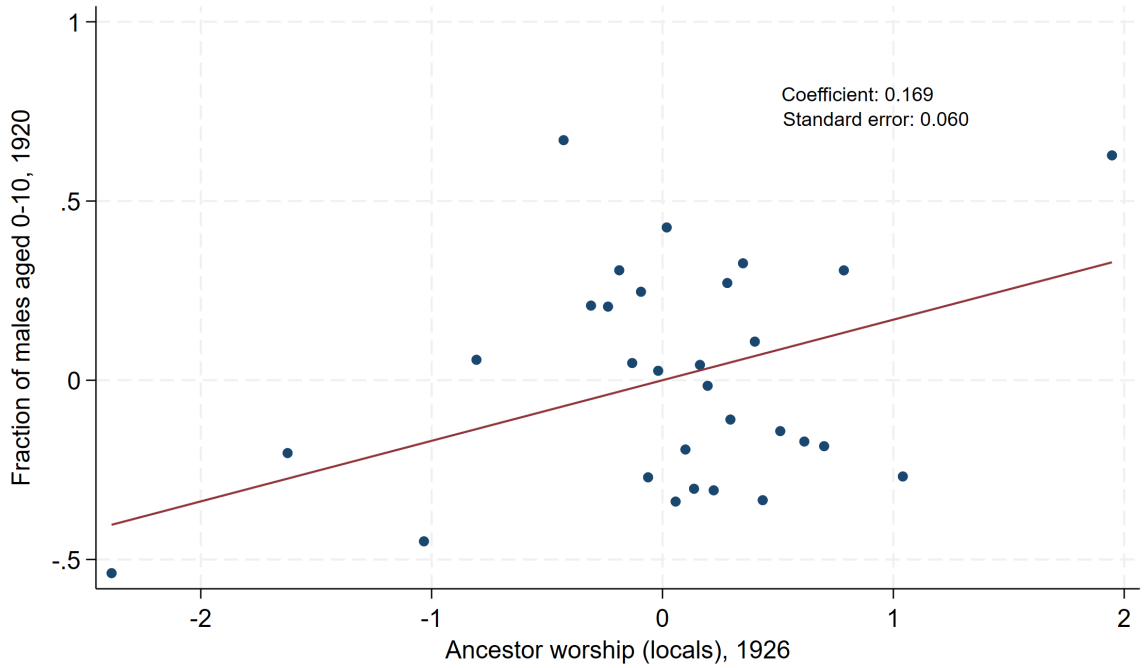

(a) Fraction of Males at Birth (Population)




(b) Fraction of Males at Birth (Locals)

Note: Trends of the fraction of males at birth and the fraction of local males at birth across groups of towns with high, medium, and low ancestor worship (migrants) in 1906-1931.

Figure A.6: Positive Correlation between Fraction of Males Aged 0-10 and Ancestor Worship (Locals)




Note: The coefficient is standardized, one s.d. increase in ancestor worship (locals) relates to 0.169 s.d. increase in the fraction of males aged 0-10 in 1920.

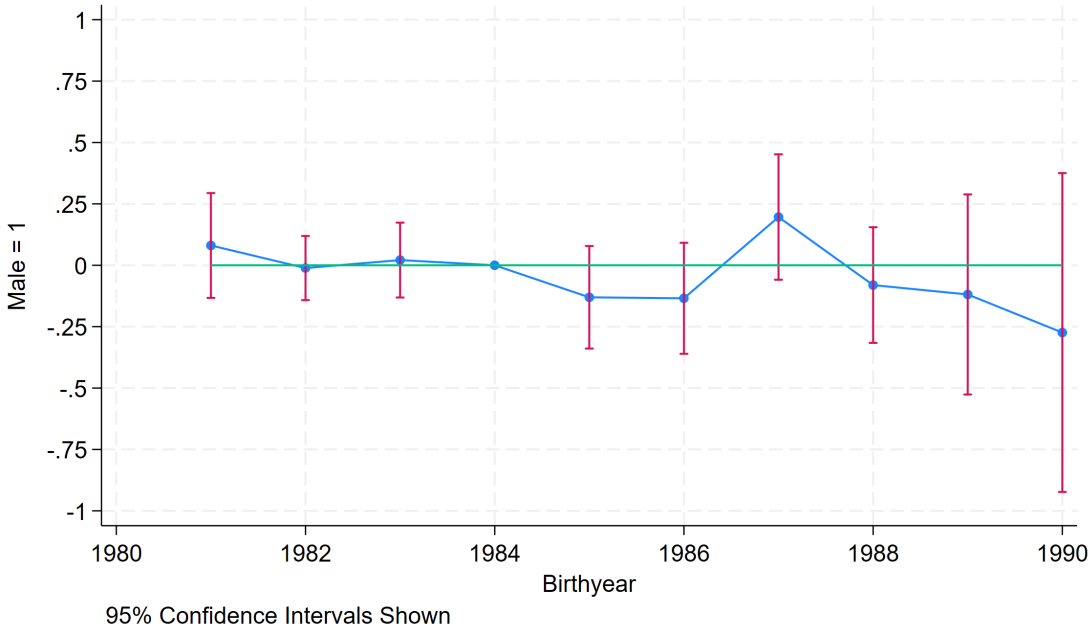



Figure A.7: Event Study for the First Two Birth Order Children (1981-1990)

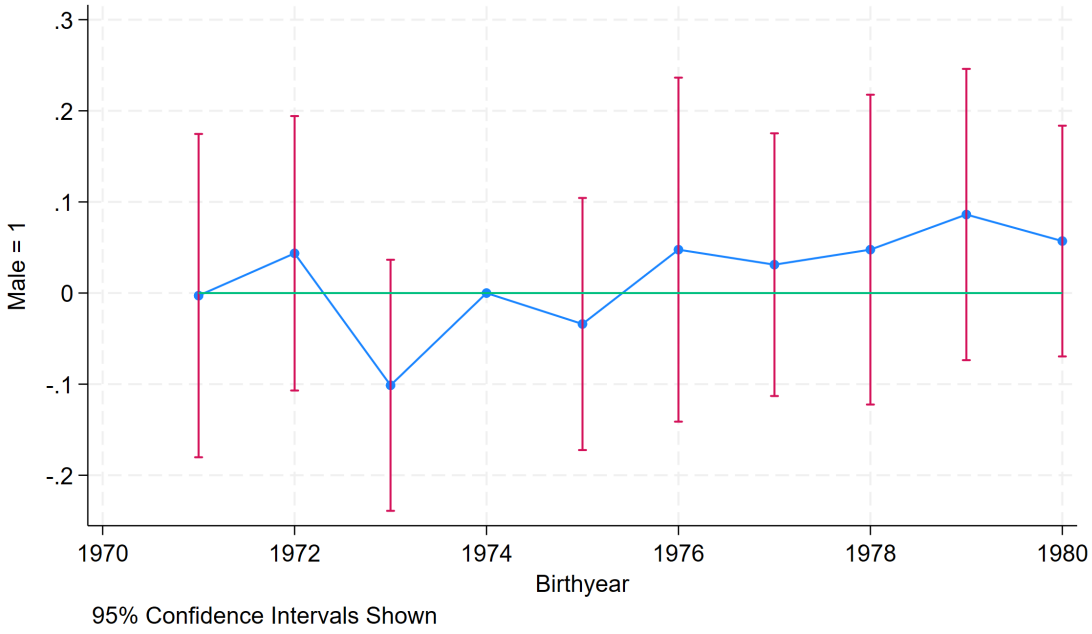



Figure A.8: Placebo Test for Children in 1980 Census (1971-1980)

Table A.1: Summary Statistics of Town Level Characteristics

| | Obs. | Mean | SD |
|---|---|---|---|
| Ancestor worship (migrants) | 354 | 0.300 | 0.0148 |
| Ancestor worship (locals) | 354 | 0.331 | 0.132 |
| Fraction of males aged 0-10, 1920 | 354 | 0.512 | 0.0143 |
| Fraction of males, 1940 | 354 | 0.508 | 0.0162 |
| Fraction of Japanese, 1940 | 354 | 0.0405 | 0.0625 |
| Population density (per $km^2$), 1940 | 354 | 3164.8 | 12599.6 |
| Fraction of males of migrants, 1956 | 354 | 0.653 | 0.0975 |
| Fraction of migrants, 1956 | 354 | 0.0614 | 0.0915 |
| Ln cotton suitability | 354 | 6.901 | 1.955 |
| Ln maize suitability | 354 | 7.277 | 1.497 |
| Ln tea suitability | 354 | 8.039 | 1.165 |
| Ln wet rice suitability | 354 | 7.627 | 1.953 |
| Ln white potato suitability | 354 | 7.157 | 1.341 |
| Ln wheat suitability | 354 | 7.350 | 1.404 |
| Ln distance to seashore | 354 | 2.699 | 0.725 |
| Ln distance to Taipei | 354 | 4.777 | 1.047 |

Table A.2: Summary Statistics of Aged 0-9 Children with 3+ Birth Order, Local Parents

| | Obs. | Mean | SD |
|---|---|---|---|
| Male | 358,287 | 0.527 | 0.499 |
| Birth order | 358,287 | 3.482 | 0.972 |
| Father's age at child's birth | 358,287 | 28.80 | 3.121 |
| Father's birthyear | 358,287 | 1958.38 | 2.962 |
| Father's year of schooling | 358,287 | 8.887 | 2.665 |
| Mother's age at child's birth | 358,287 | 26.38 | 3.327 |
| Mother's birthyear | 358,287 | 1960.96 | 3.417 |
| Mother's year of schooling | 358,287 | 7.938 | 2.545 |

Table A.3: Summary Statistics of Aged 0-9 Children with 3+ Birth Order, Migrant Parents

| | Obs. | Mean | SD |
|---|---|---|---|
| Male | 5,792 | 0.529 | 0.499 |
| Birth order | 5,792 | 3.168 | 0.452 |
| Father's age at child's birth | 5,792 | 29.65 | 3.184 |
| Father's birthyear | 5,792 | 1957.27 | 2.781 |
| Father's years of schooling | 5,792 | 11.034 | 3.045 |
| Mother's age at child's birth | 5,792 | 27.35 | 3.414 |
| Mother's birthyear | 5,792 | 1959.58 | 3.221 |
| Mother's years of schooling | 5,792 | 9.455 | 3.026 |

Table A.4: Summary Statistics of the TSCS, Local Sample

| | Obs. | Mean | SD |
|---|---|---|---|
| Son preference score | 6,084 | 0.520 | 0.382 |
| Male | 6,084 | 0.506 | 0.500 |
| Age | 6,084 | 44.38 | 15.699 |
| Birthyear | 6,084 | 1961.38 | 16.381 |
| Years of schooling | 6,084 | 10.896 | 4.760 |
| Ever-married or cohabiting | 6,084 | 0.755 | 0.430 |
| Number of children | 6,084 | 1.984 | 1.621 |

Table A.5: Migrants and Origin Characteristics in Mainland China

| Dependent variable | Log Migrant population | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Log Population of origin | 1.38*** | 1.36*** | 1.26*** | 1.27*** |
| | (0.14) | (0.11) | (0.17) | (0.17) |
| Log Population density of origin | | 38.89*** | 37.58*** | 33.83*** |
| | | (10.45) | (10.80) | (10.19) |
| Ancestor worship density | | | 2.69 | 2.20 |
| | | | (2.86) | (3.02) |
| Confucian scholars density | | | | 0.01 |
| | | | | (0.03) |
| Clan density | | | | 0.95 |
| | | | | (0.66) |
| # of Origins | 48 | 48 | 48 | 48 |
| $R^2$ | 0.620 | 0.702 | 0.709 | 0.726 |

Notes: The sample is 48 origins in mainland China, defined by KMT. This table reports coefficients from regressing log migrant population on log home population in 1953, home population density in 1953, ancestor worship density, Confucian scholars density, and clan density in 1911. Robust standard errors are in parentheses.
*** Significant at the 1 percent level.

Table A.6: The Effect of Ancestor Worship (Migrants) on the Sex of the 3+ Birth Order – Alternative Specifications

| Dependent variable | Male = 1 | | | | | |
|---|---|---|---|---|---|---|
| | County clustering | Two-way clustering | Parents' origins | Year 1985 as pre-period | Logit | Probit |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Ancestor worship (migrants) X Post | 0.29*** | 0.29*** | 0.29*** | 0.22** | 0.29*** | 0.29*** |
| | (0.07) | (0.07) | (0.08) | (0.09) | (0.08) | (0.08) |
| Parents' origins-birthyear FEs | No | No | Yes | No | No | No |
| # of Counties | 21 | 21 | 21 | 21 | 21 | 21 |
| # of Towns | 354 | 354 | 354 | 354 | 354 | 354 |
| # of Observations | 358,287 | 358,287 | 358,287 | 358,287 | 358,287 | 358,287 |
| Outcome mean | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 |

Notes: This table reports coefficients from estimating Specification (1) with alternative specifications on the sample of $3^{rd}$+ birth order aged 0-9 children whose parents are both locals. All the fixed effects and covariates are controlled in each column. Standard errors are clustered at the town level, except in column 1, where I cluster at the county level, and in column 2, where I allow for two-way clustering at the town and birthyear levels. In column 3, I include parents' origin-birthyear fixed effects to control for heterogeneous effects of family culture. In column 4, I set *Post* equal to 0 for the 1985 birth cohort. In columns 5 and 6, I report the marginal effects of Logit and Probit models at the mean of covariates.
*** Significant at the 1 percent level. ** Significant at the 5 percent level.

Table A.7: Correlation of Ancestor Worship (Migrants) with Alternative Measures Weighted by Original Population and Original Area, or Generated by Number of Ancestor Worship Festivals

| Dependent variable | Ancestor worship (migrants) | | Ancestor worship (festivals) | | |
|---|---|---|---|---|---|
| Weights | Pop | Area | None | Pop | Area |
| | (1) | (2) | (3) | (4) | (5) |
| Ancestor worship (migrants) | 1.09*** | 0.87*** | 9.29*** | 10.55*** | 7.71*** |
| | (0.02) | (0.04) | (1.09) | (1.10) | (1.38) |
| Towns (Observations) | 354 | 354 | 354 | 354 | 354 |
| $R^2$ | 0.939 | 0.913 | 0.330 | 0.316 | 0.237 |

Note: The table is to present the correlation between different statistics of ancestor worship culture, i.e. aggregating ancestor worship density and the number of ancestor worship festivals from county to province with weighted in population and area and without weights. Population data is from the 1953 China Population Census. Area data is from the China Historical Geographic Information System. Robust standard errors are in parentheses.
*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table A.8: Alternative Measures of Ancestor Worship: Weights and Numerator

| Dependent variable | Male = 1 | | | | | |
|---|---|---|---|---|---|---|
| Variable Z | Ancestor worship (migrants) | | | Ancestor worship (festivals) | | |
| Weights of Z | None | Pop | Area | None | Pop | Area |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Variable Z X Post | 0.29*** | 0.25*** | 0.31*** | 0.02** | 0.01* | 0.02** |
| | (0.08) | (0.09) | (0.08) | (0.01) | (0.01) | (0.01) |
| Outcome mean | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 |
| $R^2$ | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 |

Notes: This table replicates column 4 of Table II with alternative measures of ancestor worship culture brought by Chinese migrants. The alternative measures are mentioned in Table A.7. All the fixed effects and covariates are controlled in each column. Standard errors are all clustered at the town level.
*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table A.9: Alternative Measures of Ancestor Worship: KMT and CCP Controlled Areas before the Civil War

| Dependent Variable | Male = 1 | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Ancestor worship (migrants, KMT) X Post | 0.32*** | | 0.31*** |
| | (0.08) | | (0.09) |
| Ancestor worship (migrants, CCP) X Post | | 0.07 | 0.05 |
| | | (0.06) | (0.06) |
| # of Towns | 354 | 354 | 354 |
| # of Observations | 358,287 | 358,287 | 358,287 |
| Outcome mean | 0.527 | 0.527 | 0.527 |
| $R^2$ | 0.002 | 0.002 | 0.002 |

Notes: This table replicates column 4 of Table II with alternative measures of ancestor worship culture brought by Chinese migrants. To address possible measurement error from aggregating county level ancestor worship to the province level, the alternative measures use ancestor worship from either KMT-controlled areas or CCP-controlled areas in June 1946, on the eve of the Civil War. All the fixed effects and covariates are controlled in each column. Standard errors are all clustered at the town level.
*** Significant at the 1 percent level.

Table A.10: Horse Racing: Other Festivals in the County Gazetteers

| Dependent variable | Male = 1 | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Ancestor worship (migrants) X Post | 0.29*** | | | | 0.32** |
| | (0.08) | | | | (0.14) |
| Religious festivals (migrants) X Post | | 0.07 | | | 0.11 |
| | | (0.05) | | | (0.13) |
| Marketplace festivals (migrants) X Post | | | 0.10 | | -0.15 |
| | | | (0.11) | | (0.29) |
| Agricultural festivals (migrants) X Post | | | | 0.18 | -0.25 |
| | | | | (0.17) | (0.45) |
| # of Towns | 354 | 354 | 354 | 354 | 354 |
| # of Observations | 358,287 | 358,287 | 358,287 | 358,287 | 358,287 |
| $R^2$ | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 |
| Outcome mean | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 |

Notes: This table replicates column 4 of Table II with measures of other festivals generated from county gazetteers. The festivals are religious festivals (excluding ancestor worship), marketplace festivals (business activities), and agricultural festivals. I use ChatGPT-5 to digitize the county gazetteers and generate the share of corresponding festivals. The construction of other festival measures is similar to ancestor worship (migrants), except that ancestor worship density is replaced with the share of other festivals. The estimate in column 1 is the baseline estimate in column 4 of Table II. All fixed effects and covariates are controlled in each column. Standard errors are clustered at the town level.
*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table A.11: Horse Racing: Other Son Preference related Cultures

| Dependent variable | Male = 1 | | | | | | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Ancestor worship (migrants) X Post | 0.29*** | | | | | | 0.37*** |
| | (0.08) | | | | | | (0.11) |
| Clan (migrants) X Post | | 0.01 | | | | | -0.01 |
| | | (0.01) | | | | | (0.01) |
| Confucian scholars (migrants) X Post | | | 0.00 | | | | -0.00 |
| | | | (0.00) | | | | (0.00) |
| Tea (migrants) X Post | | | | 0.00 | | | -0.03** |
| | | | | (0.01) | | | (0.01) |
| Wet rice (migrants) X Post | | | | | 0.02* | | 0.04** |
| | | | | | (0.01) | | (0.02) |
| Wheat (migrants) X Post | | | | | | 0.02 | -0.05 |
| | | | | | | (0.02) | (0.03) |
| # of Towns | 354 | 354 | 354 | 354 | 354 | 354 | 354 |
| # of Observations | 358,287 | 358,287 | 358,287 | 358,287 | 358,287 | 358,287 | 358,287 |
| Outcome mean | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 | 0.527 |
| $R^2$ | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 |

Notes: This table replicates column 4 of Table II with measures of other placebo cultures brought by Chinese migrants. The cultures are family clan culture, Confucian culture, tea culture, rice culture, and wheat culture. The construction of these cultural measures is similar to ancestor worship (migrants), except that ancestor worship density is replaced with the corresponding cultural measures. The estimate in column 1 is the baseline estimate in column 4 of Table II. All fixed effects and covariates are controlled in each column. Standard errors are clustered at the town level.
*** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table A.12: The Heterogeneous Effect by Migrant Share and Cultural Distance

| Dependent Variable | Male = 1 | | | |
|---|---|---|---|---|
| | (1) Migrant Teacher Share | (2) Migrant Peer Share | (3) Distance to Confucian School | (4) Foot-binding Rate |
| | **Panel A. < Median** | | | |
| Ancestor worship (migrants) | -0.24 | 0.02 | 0.43*** | 0.16 |
| × Post [1] | (0.26) | (0.28) | (0.08) | (0.23) |
| # of Towns | 177 | 177 | 177 | 177 |
| # of Observations | 133,559 | 130,552 | 238,090 | 150,852 |
| Outcome mean | 0.524 | 0.524 | 0.527 | 0.528 |
| | **Panel B. > Median** | | | |
| Ancestor worship (migrants) | 0.33*** | 0.33*** | -0.00 | 0.49*** |
| × Post [2] | (0.08) | (0.10) | (0.28) | (0.10) |
| # of Towns | 177 | 177 | 177 | 177 |
| # of Observations | 224,728 | 227,735 | 120,197 | 207,435 |
| Outcome mean | 0.530 | 0.529 | 0.527 | 0.527 |
| | **Panel C. Difference between Coefficients** | | | |
| $p$-value: [1] − [2] | 0.036 | 0.297 | 0.140 | 0.188 |

Notes: This table reports OLS coefficients from estimating Specification (1) with different subsamples. Migrant Teacher Share is the share of migrants among teachers born before 1935. Migrant Peer Share is the share of migrants among residents born between 1954 and 1974. Both measures are constructed at the town level using the 1980 census. All columns contain birthyear fixed effects, town fixed effects, and a set of historical controls, as well as a series of geographic characteristics and family controls. Standard errors are all clustered at the town level.
*** Significant at the 1 percent level.

Table A.13: The Null Effect of Ancestor Worship (Locals) on Migrant Parents

| Dependent Variable | Male = 1 | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Ancestor worship (locals) X Post | 0.008 | 0.073 | 0.132 | 0.145 |
| | (0.176) | (0.176) | (0.183) | (0.178) |
| Birthyear FEs | Yes | Yes | Yes | Yes |
| Town FEs | Yes | Yes | Yes | Yes |
| Historical controls X Post | No | Yes | Yes | Yes |
| Geographic characteristics X Post | No | No | Yes | Yes |
| Family controls X Post | No | No | No | Yes |
| # of Towns | 193 | 193 | 193 | 193 |
| # of Observations | 5,792 | 5,792 | 5,792 | 5,792 |
| $R^2$ | 0.039 | 0.039 | 0.041 | 0.043 |

Notes: This table reports OLS coefficients from estimating Specification (1) on the sample of $3^{rd}$+ birth order aged 0-9 children whose parents are both second-generation migrants. The results from regressing male indicator on birthyear fixed effects, town fixed effects, and a set of historical controls, as well as a series of geographic characteristics and family controls interacting with *Post*. *Geographic characteristics:* the suitability index of cotton, maize, wet rice, white potato, wheat, and tea; (ln)distance to seashore, (ln)distance to Taipei. *Family controls:* birth order of the child, age of parents at child's birth, and years of schooling of parents. Standard errors are all clustered at the town level.

Table A.14: The Effect of Migrant Mother's Ancestor Worship on the Sex of 3+ Birth Order Children

| Dependent variable | Male = 1 | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Ancestor worship (mother) X Post | -1.88 | -3.16* | -3.15* |
| | (1.18) | (1.66) | (1.67) |
| Town-birthyear FEs | Yes | Yes | Yes |
| Mother's origin FEs | Yes | Yes | Yes |
| Father's origin-birthyear FEs | Yes | Yes | Yes |
| Origin controls (mother) X Post | No | Yes | Yes |
| Family controls X Post | No | No | Yes |
| # of Mother's origins | 32 | 32 | 32 |
| # of Towns | 192 | 192 | 192 |
| # of Observations | 5,753 | 5,753 | 5,753 |
| Outcome mean | 0.529 | 0.529 | 0.529 |
| $R^2$ | 0.220 | 0.221 | 0.223 |

Notes: This table reports coefficients from estimating Specification (2) on the sample of $3^{rd}$+ birth order aged 0-9 children whose parents are both second-generation migrant . The results from regressing male indicator on town-birthyear fixed effects, mother's origin fixed effects, father's origin-birthyear fixed effects, and origin controls (mother) as well as a series of family controls interacting with *Post*. *Origin controls (mother):* Imperial scholars density, Confucian clan density, and suitability index of wheat, wet rice, and tea of mother's origin. *Family controls:* birth order of the child, age of parents at child's birth, and years of schooling of parents, as well as whether parents are from the same origin. Standard errors are all clustered at the town level and migrant mother's origin level.
* Significant at the 10 percent level.

Table A.15: The Effect of Ancestor Worship of Migrant Neighbors from Other Origins than Mother on the Sex of 3+ Birth Order Children

| Dependent variable | Male = 1 | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Ancestor worship (neighbors, mother) X Post | 4.98* | 3.52 | 4.12 |
| | (2.75) | (2.99) | (2.97) |
| Town-birthyear FEs | Yes | Yes | Yes |
| Parents' origin-birthyear FEs | Yes | Yes | Yes |
| County-Mother's origin FEs | No | Yes | Yes |
| Family controls X Post | No | No | Yes |
| # of Mother's origin | 26 | 26 | 26 |
| # of Towns | 186 | 186 | 186 |
| # of Observations | 5,558 | 5,558 | 5,558 |
| Outcome mean | 0.528 | 0.528 | 0.528 |
| $R^2$ | 0.239 | 0.286 | 0.290 |

Notes: This table reports coefficients from estimating Specification (3) on the sample of aged 0-9 children whose parents are both second-generation migrants. The results from regressing male indicator on town-birthyear fixed effects, parents' ancestry-birthyear fixed effects, and a series of family controls interacting with *Post*. *Family controls:* birth order of the child, age of parents at child's birth, and years of schooling of parents, as well as whether parents are from the same origin. Standard errors are clustered at the town level and migrant mother's origin level.
* Significant at the 10 percent level.